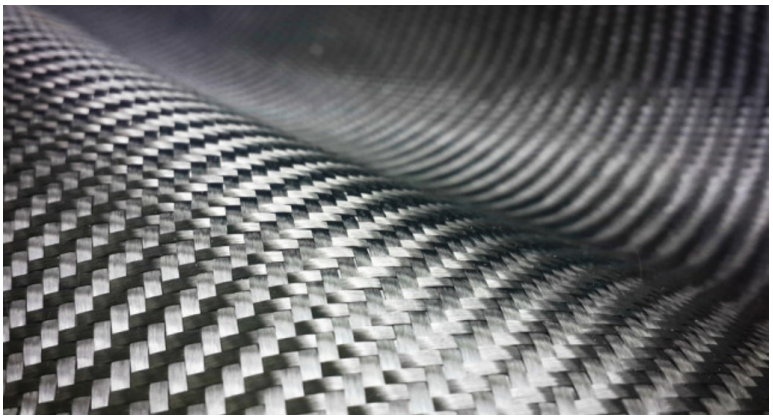
MOTORVEHICLE
UNIVERSITY OF
EMILIA-ROMAGNA

Teamwork Report

A.Y. 2023/24



Design of Racing car composite structures: laboratory activity

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1 DSC report

1.1 Theory

Differential scanning calorimetry (DSC) testing is a pure thermal technique that allows to evaluate the matrix properties of the sample, without any contribution of fibers, being negligible in the temperature range of interest (below the degradation temperature). It belongs to general group of the "thermoanalysis", a generic name used to describe a series of analytical techniques which measure physical and chemical changes in material as a function of temperature and time. In particular the DSC aims at measuring the difference in the amount of heat required to increase the temperature of a sample and of a reference.

DSC is based on "heat flux" measurements by scanning from low to high temperatures, in order to study the thermal and thermodynamic properties of the material, associated to material transitions. The term "differential" suggest that this technique is not based on a direct measurement but a direct signal is compared with two or more reference sensors; this is to guarantee a low signal-to-noise ratio, by cancelling the common noise between signals.

The particular aim of a DSC test is to evaluate the thermal properties of a cured or fresh piece of composite material, in order to determine the degree of cure and the glass transition temperature of the part under investigation. From the degree of cure we can define the amount of the reacted and unreacted material, that is fundamental in determining the final mechanical properties of the component, while from the Tg we can identify the range of temperature beyond which the material starts to soften its mechanical properties, passing from a glassy to a more rubbery state. Moreover, the resulting thermogram allows to get further information about the matrix composition from the time-temperature evolution of the curve, that can identify multiple reactions and thus components.

1.2 Laboratory objective and materials

The objective of the experimental activity is to determine the thermal properties of virgin prepregs CFRP samples, by means of the Differential Scanning Calorimetry technique. The thermal characterization is performed by evaluating:

- qualitative detection of material transitions and their nature (reversible or irreversible)
- Glass transition temperature (T_g)
- Curing temperature (T_c)
- Enthalpy of reaction (H)
- Degree of polymerization $\alpha = \dfrac{H_R - H}{H_R}$, where H is the enthalpy of reaction of a partially cured sample, while H is the one of complete fresh material, obtained by a full cure of both of them.

In particular two different prepreg materials have been analysed, each one with specific characteristics provided by the datasheets: T800 ER450 38% and M46J ER434 38%

T800 ER450 38%: T800 carbon fibre pre-impregnated with a ER450 resin, in 38% of weight. T800 is a never twisted fiber with excellent tensile composite properties but intermediate modulus and is specifically designed to meet the weight saving demands.

	M46J	T800
Tensile Strength [Mpa]	4000	5490
Tensile modulus [Gpa]	436	294
density [g/cm^3	1.84	1.80

	ER450	ER434
Max service temperature	180°C	125°C
Temperature of cure	80 – 180	85 -135
Typical curing time	120 min at 135°C	90 min at 125°C
Expiration time	Up to 45 days shop life at room temperature	Up to 30 days shop life at room temperature

Table 1: technical data for fibers and matrices

ER450 is a toughened epoxy resin developed specifically for a wide range of structural applications in the automotive and aerospace industry and suitable for high temperature application (maximum service temperature 180°C). It can be stored at room temperature up to 45 days, without suffering any "aging" effect, and it is characterized by a relatively long curing cycle (2h at 135°C).

M46J ER434 38%: M46J carbon fiber is an MJ-type high modulus fiber with enhanced tensile and compressive strength. ER434 is a structural resin with high toughness and impact resistance. It is characterized by a lower maximum service temperature (125°C) and a lower storage time at room temperature (30 days). It is characterized by a lower curing time (typically 90 min at 125°C)

1.3 Test description

The preparation of the sample is an important phase of the test, that must be consistent with the indication provided by the standard chosen to perform the analysis. In this case a specimen of a virgin material has to be cut in order to characterize its properties. So, first of all it is necessary to cut, with some scissors, a small piece of material that will represent the sample to analyse: the standard ASTM D3418 requires to use samples, generally not larger than 20 mg. The samples must be weighted on a high precision balance (sensitivity 10^-5 grams).

For this activity the respective masses of the samples are reported below:

	T800 ER450 38%:	M46J ER434 38%
Sample mass	21.27 mg	9.110 mg

Table 2: Hardpoints table and DOE variations

Once cut, the sample must be placed into an empty aluminum crucible of 40 microliters. To obtain the actual weight of the sample the crucible mass has to be subtracted from the total weight.

After that, the crucible is closed and sealed with a pressing pin to prevent the liquid resin going on the sensor. Finally the sample is placed into the machine test chamber and the processing parameters are set on the software to perform the test: initial temperature, rate of temperature change and final temperature.

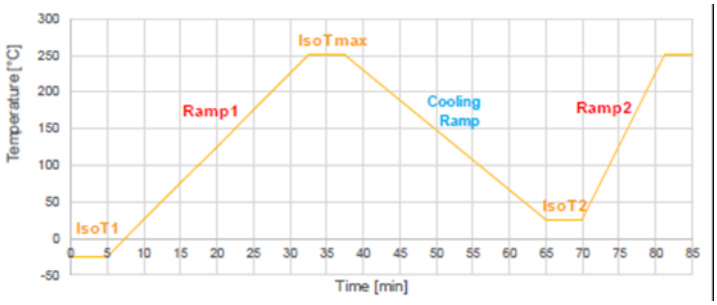


Figure 1: Caption

Last but not least, a reference material (a constantan disk) is placed in another crucible to characterize the difference in heat flow between the sample and the reference. Before performing the thermal characterization of the material it is necessary to communicate to the control temperature system of the machine how to realize the scan of temperature. According to the standard ASTM E2160, the following temperature steps has to be performed (see also Fig. [1])

- **Isotherm -T1** = -25°C for 5 min to stabilize the starting temperature of sample and reference; starting from such low temperature the material will surely start from a temperature below the Tg.
- **Ramp1** from T = -25°C to T = 250°C at 10°C/min rate as stated by the standard, during which the material cure is performed; the max temeprature is set at 250°C to stay higher than the curing temperature provided by the datasheet, but lower than the degradation temperature.
- **Isotherm-T**$_{max}$ = 250°C for 5 min; to stabilize the final temperature
- **Cooling ramp** from T = 250°C to T = 25°C at -10°C/min rate
- **Iso-T2** = 25°C for 5 min
- **Ramp2** from T = 25°C to T = 250°C at 20°C/min (according to standard ASTM D3418), to measure the max Tg of the fully cured material

During the heating the virgin material gets cured through an exothermal reaction that raises the sample temperature. The system measures the difference in heat flux necessary to keep the constantan disk at the same temperature of the sample. Basing on this, it returns a thermogram with a typical bell curve of the polymerization process from which we can derive many parameters:

- The specific/normalized enthalpy of reaction, deduced from the area below the peak ΔH [J/g]
- the glass transition temperature (Tg)
- the temperature of cure (Tc)
- in some cases also the degree of cure (α), when analysing partially cured component

The resulting thermogram of the cured sample will display a more flat curve, with no more the typical bell shape of uncured specimen, from which the max Tg of the polymerized part can be derived in correspondence of the flex.

1.4 Result analysis

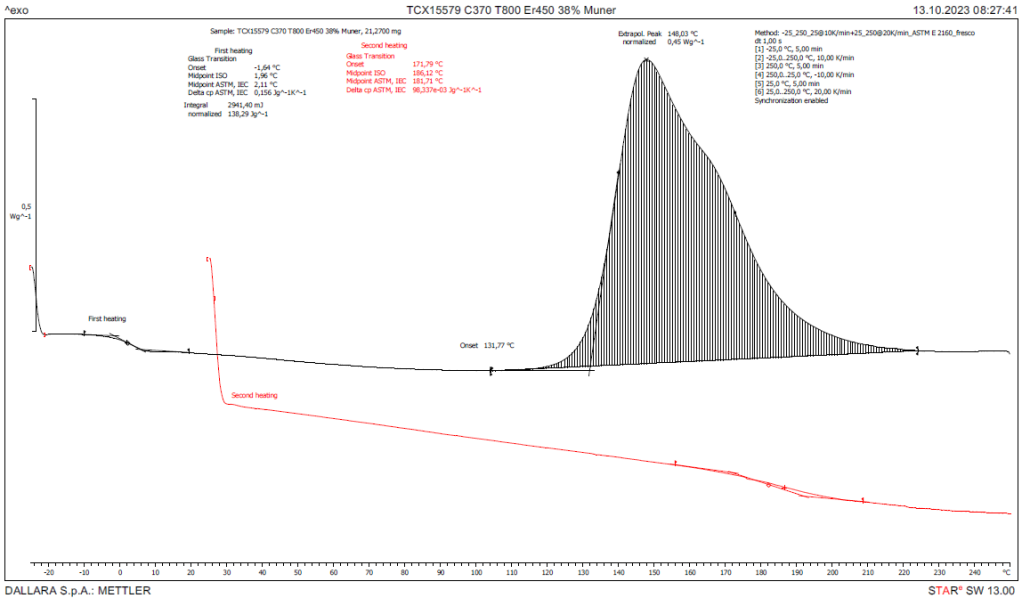


Figure 2: T800 ER450 38% - DSC CURVES (2023)

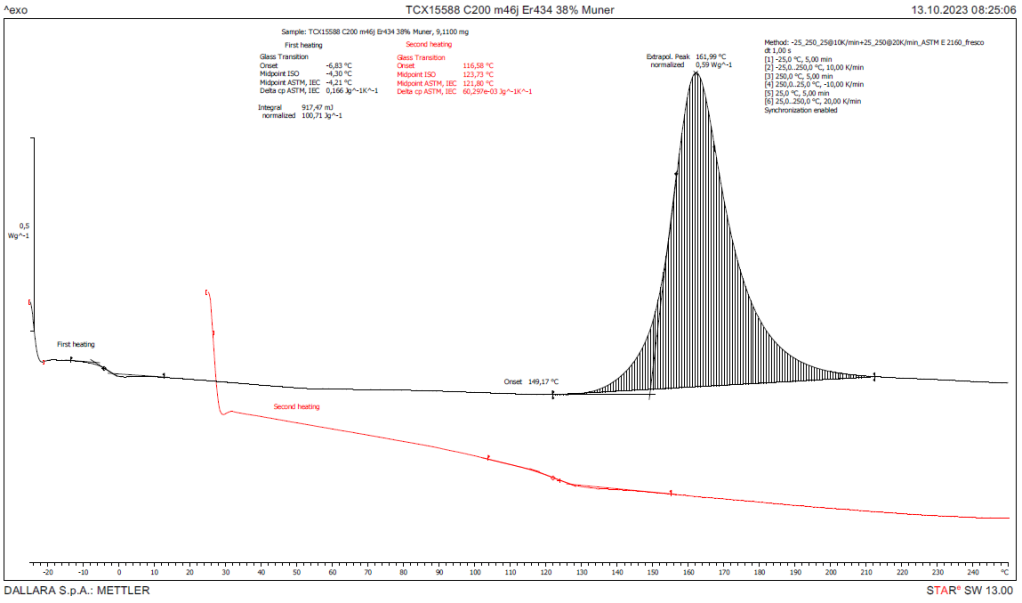


Figure 3: M46J ER434 38% - DSC CURVES (2023)

T800 ER450

Looking at the the first thermogram, reported in Fig. [2], we can state that ER450 is a quite high-Tg material (181,71°C, coherent with the datasheet) , therefore also suitable for high

temperature working conditions. Moreover, the virgin material's bell curve shows a kind of shoulder, with respect to the one of the ER434 in Fig. [5], which can be due to an activation of a second constituent inside the composition of the resin, which activates at higher temperature than the first constituent. Combining two or more components inside the resin such that they start creating links at different temperatures is usually done for increasing the properties of the matrix, like the Tg of the cured resin or some mechanical properties. This explains such a high Tg and the high enthalpy of reaction.

By the second temperature ramp (red thermogram) we can see that no further exhotermal reactions occured, meaning that after th first temeprature ramp the material gets fully cured. Moreover, during the curing process, the glass transition temperature moves from low to high temperatures, up to its maximum value at full cure, as a consequence of the more rigid structure that can be obtained by the increasing interconnected cross-linked structure.

Since, as the degree of cure increase, the peak of the exhotermal reaction collapse on the final T_g, we can extrapolate the temperature of cure as a temperature 10-20°C lower than the final T_g. However, information about the temperature and time dependence of the curing process will allow to define more precisely the paramaters affecting the polymerization process.

- Glass transition of the virgin material: T_g=1,36°C
- Glass transition of the cured material: T_g=181,71°C
- normalized enthalpy of reaction: $\frac{2941{,}40mJ}{21{,}27mg} = 138,29\frac{J}{g}$
- Temperature of cure: $T_c \simeq$ 160-170°C

M46J ER434

Looking at thermogram of the the sample M46J ER434 (Fig. 5), we can state that this time we are dealing with a lower Tg material, since its final value of the cure materuial is 121.80°C. Moreover, as already said, the bell curve does not present any shoulder, but only a single peak, thus suggesting the presence of no other constituents. Also this time the Tg increase with the degree of cure.

- Glass transition of the virgin material: T_g=-4,21°C
- Glass transition of the cured material: T_g=121,80°C
- normalized enthalpy of reaction: $\frac{917{,}47mJ}{9{,}11mg} = 100,71\frac{J}{g}$
- Temperature of cure: $T_c \simeq$ 100-110°C

1.5 2023-2022 results comparison

Thanks to the availability of old results, obtained a year before, we can verify the reliability of the results. In particular the data available from the past year are about a T300 ER434 38% and a M46J ER434 – 38% prepregs. However, even though different fibers for one of the two samples were impregnated by the same resin of the specimens analysed this year, we expect to obtain more or less the same thermal properties, apart from differences coming from a not exact content of resin, proving that the fibers does not affect the characterization or the material during this technique.

Comparing the two T800 ER450 thermograms (Fig. 2 and 4) we can see the same behaviour of the exhotermal curve, charcaterized again by a shoulder, even if less prominent, suggesting the presence of a further reacting component.

T800ER450 38%	$\mathbf{T_g^v}$ **[°C]**	$\mathbf{T_g^c}$ **[°C]**	ΔT_g^v **[°C]**	ΔT_g^c **[°C]**
2022	1.36	1183.03	0.75	1.32
2023	2.11	181,71		

Table 3: Direct comparison between 2023 and 2022 results, with differences in abasolute value

Comparing the data, we can see very small differences in the glass transition temepratures, which can be owing to a different percentage of resing in the weighted prepreg sample, leading to a different number of cross-links and thus a differen Tg.

No detailed data can be provided for the specific enthalpy of reaction because of the lack of information about the mass of the 2022 sample.

For the second couple of samples, different fibers were used, but becuase of the low temperature range no contribution of the fibers is expected, and therefore similar results are expected to be obtained.

(_ _) - ER434 38%	$\mathbf{T_g^v}$ **[°C]**	$\mathbf{T_g^c}$ **[°C]**	ΔT_g^v **[°C]**	ΔT_g^c **[°C]**	**$ H \{J}{g}**	**$\Delta H \{J}{g}**
2022	-3.51	119.98	0.70	1.82	**124.17**	**23.46**
2023	-4.21	121.80			**138.29**	

Table 4: Direct comparison betweenT800 ER434 2023 and T300 ER434 2022 results, with differences in abasolute value

Again, for what concerns the glass transition temperature, despite the differences in resin content the final Tg obtained after are almost the same. This confirms the fact that fibres are inert in DSC tests. Regarding the enthalpies of reaction, a bigger difference is detected, probably because of the higher amount of reacting material in the 2022 sample (9,47 mg vs 9,11 mg) that increased the overall temperature of the sample and so the energy released.

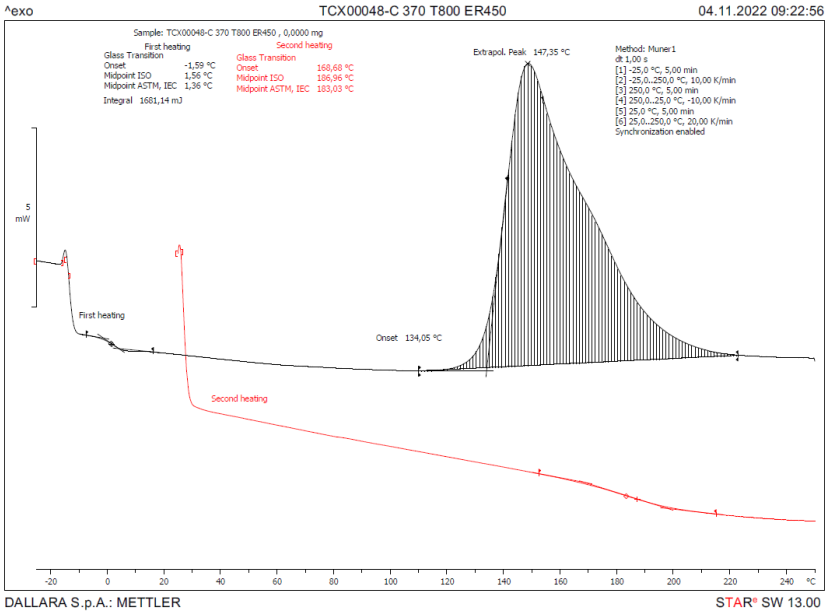


Figure 4: T800 ER450 38% - DSC CURVES (2022)

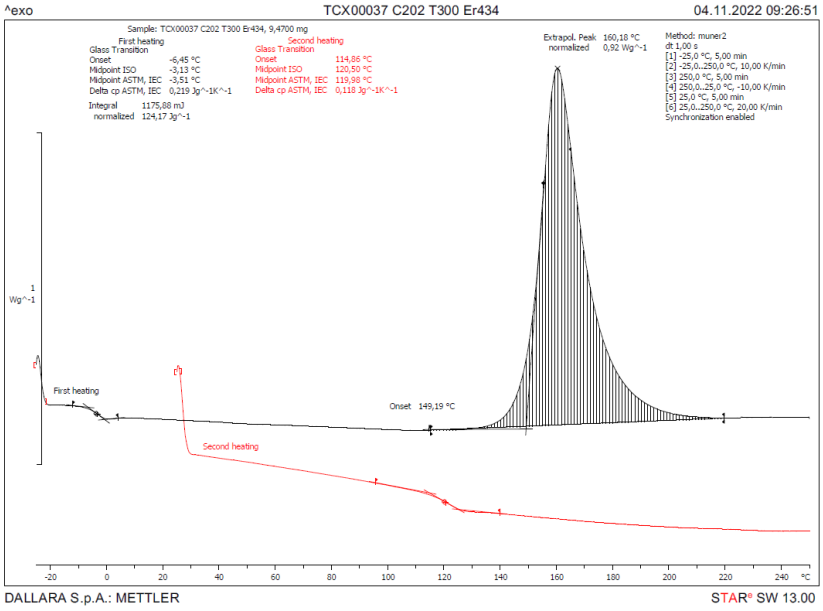


Figure 5: T300 ER434 38% - DSC CURVES (2022)

2 DMA report

2.1 Theory

Dynamic mechanical analysis (DMA) is a thermomechanical analysis that allow to characterize the mechanical and thermal properties (Tg) of a composite material by investigating how mechanical properties react to temperature stimuli. In particular, allows to determine the glass transition temperature of the matrix by analysing how the mechanical behaviour of the composite material change with temperature. To do that an oscillating force is applied, while measuring the resulting displacement, in order to evaluate the phase shift between stress and strains, while applying at the same time a temperature ramp. According to the relationship between stress and strain, it is possible indeed to identify different two main kind of mechanical behaviours:

- Hookean behaviour: it's a pure elastic behaviour characterized by a linear relationship between stress and strains ($\sigma \propto \varepsilon$), which is therefore characterized by a phase shift $\delta = 0$ ((stress and strain are in phase))
- Viscous behvaiour: mechanical behvaiour charcaterized by a linear relationship between stress and strain rate, which is therefore charcaterized by a phase shift $\delta = 90$ (stress and strain are in anti-phase)

Real material shows an hybrid behaviour (visco-elastic behaviour), which can be qualified by introducing a complex elastic modulus.

$$E* = E' + iE''$$

In particular the viscoelastic behavior of the component is evaliuated with the help of the following parameters:

- Storage modulus E': takes into account reversible phenomena inside the material (elastic behavior).
- Loss modulus E'': takes into account dissipative phenomena causing energy loss (viscous behavior).
- δ: it's the lag between applied stress and measured strain signals, it allows to define the **damping** as $\tan\delta = \frac{E''}{E'}$.

The real material strain is given by the contributions of reversible and dissipative phenomena:

$$\varepsilon(t) = \varepsilon_0 \sin\omega t + \delta = \varepsilon_0 \sin\omega t \sin\delta + \varepsilon_0 \cos\omega t \sin\delta = E' \sin\omega t + E'' \cos\omega t \tag{1}$$

Depending on the material and on the behavior we are interested in, DMA can be performed in different ways:

- It is possible to apply the oscillating load with a constant amplitude and a varying frequency or viceversa with a varying amplitude and a constant frequency.
- Different test geometries can be employed, depending on the deformation mode on which we are interested, however not all of them are suitable for composites.



Figure 6: Possible DMA testing geometries.

- The test can be conducted at a specified or gradually increasing temperature, with a concurrent application of load; this allows to perform tests such as creep and stress relaxation. In a more general case the load is applied in an oscillatory manner, either maintaining a constant frequency or a constant amplitude, as the temperature ranges from room temperature to the glass temperature. As the temperature increases, the material's behavior undergoes changes, necessitating a reduction in the applied stress.

2.2 Standards

During the course we have analyzed two different kinds of standard:

ASTM D5418-15: To evaluate viscoelastic properties of thermoset or thermoplastic resins and their dynamic mechanical properties while applying flexural loading. A dual cantilever beam configuration is indeed applied together with a temperature ramp of $2°C/min$, and frequency in a $0,01 \div 100Hz$ range.

ASTM D7028-17: performed in a three points bending configuration with a temperature increase of $5°C/min$, and a frequency of $1Hz$. As a result we get storage modulus, loss modulus and $\tan\delta$ as a function of the temperature. Thus we are able to retrieve the T_g in three different ways as: peak of $\tan\delta$, onset of E', peak of E''. No information on the curing percentage is provided.

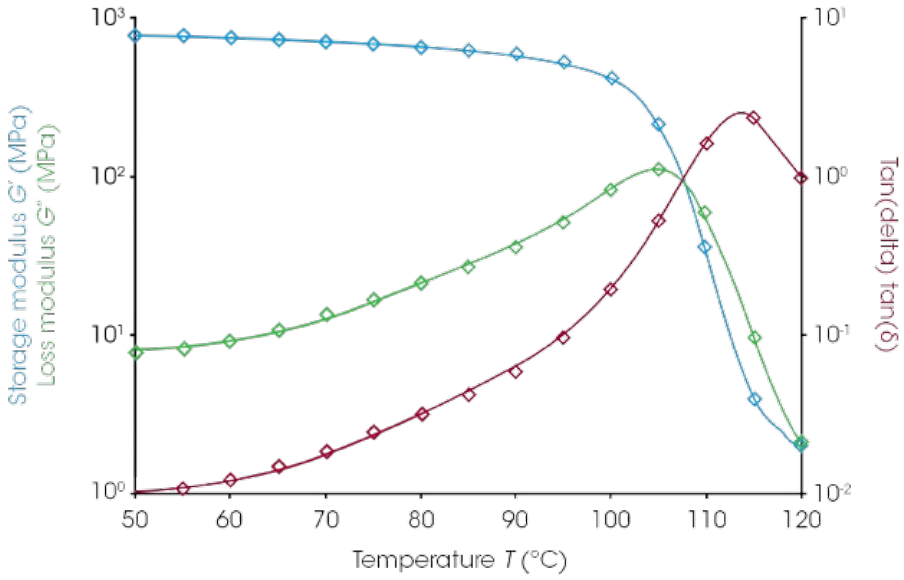


Figure 7

2.3 Laboratory activity

During our laboratory activity we have followed the ASTM-D7028-17 standard. Two kinds of laminate have been tested: a T300 ER434 specimen with 0° fibers orientation and a T800 ER450 with 90° fibers orientation.

Instrumentation: a TA Q800 CMT machine is used, composed by a furnace (collocated on the top) providing heat and thermal insulation, an actuator moved by an electric motor (on the bottom) to apply the desired load type and by control electronics. Moreover temperature is controlled by two thermocouples, one for the sample and the other for the overall furnace. A computer is then connected to the machine via an Ethernet cable, two software are used: one for the real-time dialog window and the other for the final analysis.

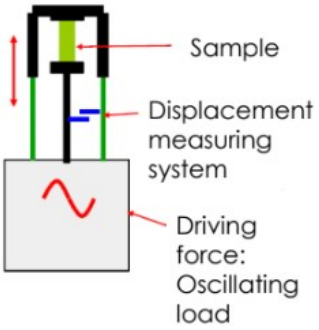


Figure 8

Calibration and set-up: calibration is a fundamental step in DMA due to the high sensitivity of the instrument to testing conditions, such as pressure and temperature. It is done using a polycarbonate sample.
First we zero the LVDT (linear variable differential transformer), then, to ensure that the sample is centered, we perform position calibration. Furthermore we carry out the

clamp calibration to measure its mass and compliance (considering the influence of the stiffness of the sample). At this point we have to measure the sample to characterize its geometrical and inertial properties.

Testing procedure: DARC performs tests following a customized procedure starting from ASTM-D7028 with a three points bending clamp (with a $60mm$ long sample), a testing frequency of $1Hz$ and, starting by maintaining $35°C$ for $5min$, it was then ramped to T_{max} (about $250°C$) with $5°C/min$ rate[1].

Result Analysis: since we are analyzing a matrix dominated test typology, it is of interest to have an overview of the epoxy resins under study:

Property	Unit	Value	Standard
Storage life @ -18°C (0°F)	months	12	
Out life @ 23°C (73°F)	days	30	
Prepreg volatiles	%wt	<1	ASTM D3530-97R03
Cured resin density	g/cm³	1.19	ASTM D792-00
Tg (DSC)	°C (°F)	130 (266)	ASTM D3418-03
Tg E' (DMA)*	°C (°F)	126 (259)	ASTM E1640-09
Tg Peak Tan δ (DMA)*	°C (°F)	137 (279)	ASTM E1640-09
Tack		Medium	

*Laminate Cured 90'@125°C

(a)

MAIN CHARACTERISTICS

- Maximum Tg of 130°C (266°F).
- Toughened epoxy matrix
- High compressive properties
- Suitable to impregnate a very wide range of support (UD, fabrics and multi-axial - carbon, glass, aramid and hybrids).
- Versatile in different manufacturing process and wide range of temperatures.

(b)

Figure 9: ER434 general properties and main characteristics from the CIT (Composites materials Italy) website.

Property	Unit	Value	Standard
Storage life @ -18°C (0°F)	months	12	
Out life @ 23°C (73°F)	days	45	
Prepreg volatiles	%wt	< 1	ASTM D3530-97R03
Cured resin density	g/cm³	1.23	ASTM D792-00
Tg (DSC)	°C (°F)	181 (358)	ASTM D3418-03
Tg E' (DMA)*	°C (°F)	171 (333)	ASTM E1640-09
Tg Peak Tan δ (DMA)*	°C (°F)	187 (363)	ASTM E1640-09
Tack		medium	
Max Dry Tg (DMA)**	°C (°F)	180 (356)	ASTM E1640-09
Max Wet*** Tg E' (DMA)	°C (°F)	120 (248)	ASTM E1640-09

* Laminate after cure at 135°C for 2hours.
** Laminate after cure at 180°C for 2hours.
*** Laminate saturated at 85°C / 85%R.H.

(a)

MAIN CHARACTERISTICS

- Flexible cure: 80÷180°C
- Maximum Tg of 210°C (410°F).
- Maximum service temperature
- of 180°C (356°F).
- Autoclave, press or vacuum bag processing.
- Toughened epoxy matrix.
- Solvent free.
- Available on a wide range of reinforcements (carbon, glass, aramid and hybrids, UD, fabrics and multi-axial).
- Mechanical properties suitable for structural requirements.
- Up to 45 days shop life at room
- temperature.
- Controlled flow system.

(b)

Figure 10: ER450 general properties and main characteristics from the CIT (Composites materials Italy) website.

[1]It has to be considered the impact of the cooling system limitations (no coolant) on test conditions, as the starting conditions for each test may be influenced by the prior use of the machine.

T300ER434 test: from the first test a storage modulus $E' = 110,7GPa$ and a glass transition temperature $T_G = 109.21°C$ (computed in correspondence of the onset of E') are obtained, then since there is a drop of properties the test ends at about $160°C$.

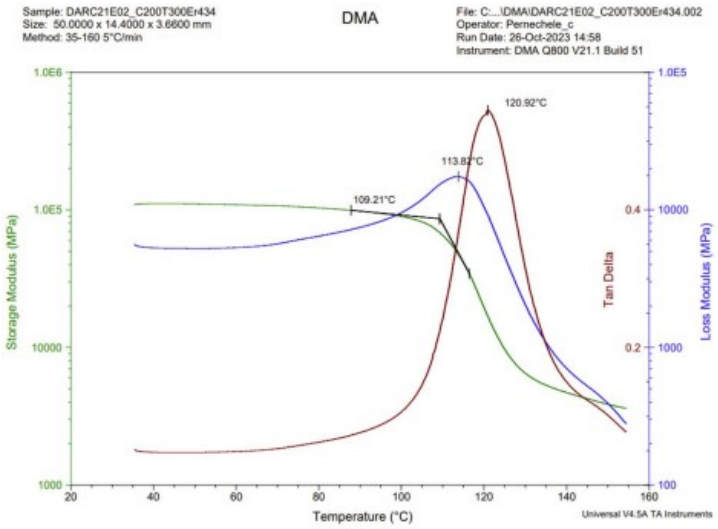


Figure 11: DMA7028-17 test for the T300ER434.

T800ER450 test: in the second test instead a slightly lower storage modulus $E' = 87,5GPa$ is obtained, on the other hand we have an higher $T_G = 139°C$. The test is carried on until the final temperature of $250°C$.

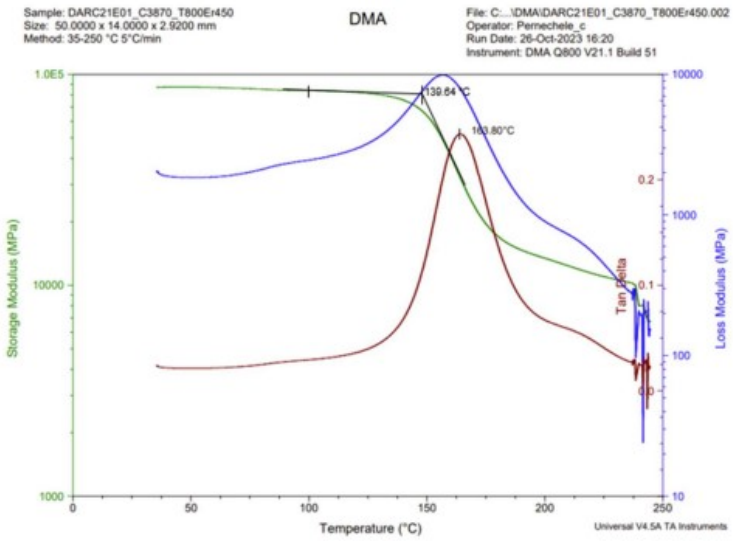


Figure 12: DMA7028-17 test for the T800ER450.

By observing the T_G obtained for the two materials we can state that, coherently with the properties of the epoxy resins used (figures (a) and (a)), we could employ the T300ER450 for higher temperature applications. The 7800ER450 instead shows higher mechanical properties from room temperature to about $100°C$.

2.4 2023-2022 Results Comparison

By comparing our results with the ones obtained during the previous year course we can notice some variations.

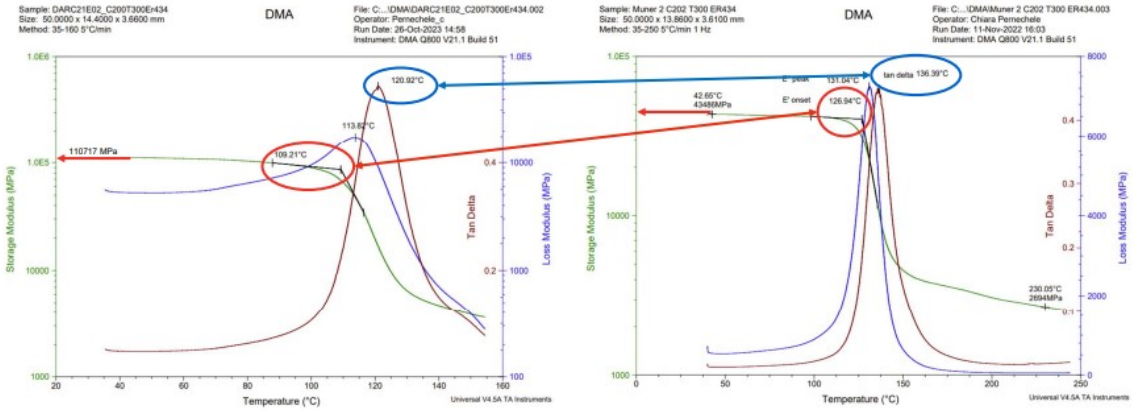


Figure 13: Differences between the tests done in 2022 and in 2023 for the T300ER434.

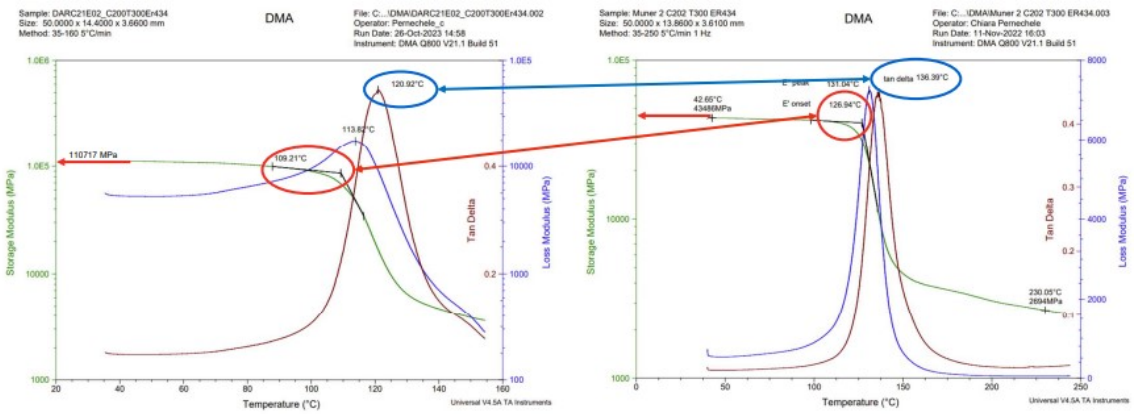


Figure 14: Differences between the tests done in 2022 and in 2023 for the T800ER450.

It can be noticed that in both cases we have more or less the same differences: the 2023 materials present a T_G which is lower of about $20°C$, on the other hand E' is higher of about $40GPa$. Probably this is due to a variation of the curing cycle of the material made with the purpose of slightly reducing the maximum working temperature in order to increase the mechanical properties.

Furthermore the 2022 test for the T800ER450 presents a strange behavior, indeed the drop of E' presents two different phases with different slopes, this could be due to a portion of the material with an higher T_G.

To conclude in the table below are provided some informations that allow to compare DSC and DMA.

DSC	DMA
fresh material (unpolymerized)	destructive testing
pure thermal technique	sample preparation necessary
inhert fibers	only polymerized material
short time required	geometry of the sample is crucial
small mass (10/20mg)	thermo-mechanical technique

3 Tension and compression report

3.1 Theory

Tensile and compression tests belong to the family of mechanical testings, thus, with respect to thermal (DSC) and thermomechanical (DMA) analyses, allow to determine only the mechanical properties of composite materials, which are mainly fiber dominated properties, even if the matrix contribution is not completely negligible. In particular, in tension the fiber plays the most important role in defining the tensile and stiffness properties of the material. In case of compression instead, because of the fiber instability to this kind of load, the matrix is more relevant, due to its function keeping fibers together, contributing to the compressive strength of the laminate.
Moreover, since the tests are performed on laminates and not on single plies, the resulting properties will be affected not only by the constituents characteristics (according to the rules of mixture) but also by the preparation and stacking sequence of the laminate. This kind of tests allow to define the following material properties, where the orthotropic and asymmetric nature to tension and compression is taken into account.

- Tension:
  - Longitudinal and transverse tensile strength
  - Elongation at break $\varepsilon_f = \frac{(l_f - l_0)}{l_0}$;
  - Elastic moduli (Longitudinal, transversal and shear modulus):
    $E_{11} = E_f V_f + E_m V_m \qquad E_{22} = \frac{(E_f E_m)}{(E_f V_m + E_m V_f)}$
    $G_{12} = \frac{(G_f G_m)}{(G_f V_m + G_m V_f)}$
  - Poisson ratios: $\nu_{12} = \nu_f V_f + \nu_m V_m \qquad \nu_{21} = E_{22}/E_{11}\nu_{12}$
- Compression
  - Longitudinal compressive strength
  - Longitudinal and transverse elastic modulus
  - transverse compressive strength

The geometry of the sample, in case of tensile test, is defined according to the lamination sequence: for quasi-isotropic samples a dogbone shape is used, while for orthotropic layups a rectangular shape is prescribed by the standard. In particular, unidirectional (UD) samples are employed with fiber orientations either at 0° or 90°, to define longitudinal and transverse properties. Specifically in case of 90° orientations, a larger number of plies is required to compensate the lower performance. A 45° oriented laminate is instead tested to evaluate the shear elastic modulus and the in-plane shear strength.
In case of compression the sample geometry is similar but the free length is shorter in order to avoid as much as possible instability issues.
Moreover, tabs can be employed through bonding or kept in place through friction by abrasive materials between the tab and the coupon. Tabs, which are usually made of a relatively softer material (Al, Glass fiber) to better distribute the load (sometimes using tapered angles to avoid stress concentration effects), are used to avoid damage of the sample caused by the clamping machine grips.

Furthermore, in order to obtain the best experimental data it is important to be aware of all the different aspects that can affect the results and bring undesired errors:

- **stochastic errors** can be taken into account testing the highest number possible of samples to rely on a good reliability of the results of the experimental campaign. If a large variation is detected, it may be a sign of errors in the manufacturing or sample preparation.
- **material affecting factors** involve material defects, porosity and lay up preparation that degrade the mechanical properties, causing a different stress distribution from the expected one
- **coupon affecting factors**: when geometry and tolerances, prescribed by the standards, are not respected, or when the environment effects on which the material will going to be used are not considered (conditioning)
- **test affecting factors**: errors due to fixtures and speed of the test different from the ones prescribed by the standard or misalignment of the load

3.2 Instrumentation

A typical tensile testing machine consists of a load cell, cross head, extensometer, specimen grips, electronics and an actuator. Two possible machines for testing exist:

- electro-mechanical: it consists in a moving cross head actuator, driven through never ending screws inside of the columns of the structure, to impress the required displacement on the sample. However it is limited in the frequency of the applicable load to the sample, and for that reason are generally devoted to static or quasi-static testing.
- hydraulic: the actuator is placed on the lower part of the machine while the cross head is fixed. This type of machine is particularly suitable for dynamic loading.



Figure 15: Hydraulic (left) and electromechanical (right) Testing Machine

For the laboratory experience an electro-mechanical machine was used, that, for standardized tests, has a load capacity of 100kN. Such a machine, in particular, is made up of:

- **Rigid frame**: column chassis rigid enough to remain undeformable under the load applied to the sample. In case of a high-capacity testing machine, we have the option to add a climatic chamber for conducting not only mechanical tests but also standard thermo-mechanical tests if needed.

- **The moving crosshead**: it is essentially a crosshead that is controlled to move up or down. In a tensile test, the crosshead speed of the testing machine is directly related to the strain rate in a specimen.
- **Load cell**: it converts the tensile force into an electrical signal that can be measured.
- **LVDT displacement transducer**: It allows to measure the displacement of the sample with respect to the movement of the cross head, where it is usually embedded. However, for this reason the compliance of the cross head contributes to the measure, therefore an external independent device may be necessary (extensometer).
- **extensometer**: device, attached externally to the sample, used to measure only its axial displacement. Based on the need, it may be axial or biaxial. In case of a biaxial extensometer the possibility of measure the displacement in two orthogonal directions allows to compute also the Poisson ratio of the material.
  However, since the extensometers can be damaged during the failure of the sample, a good alternative is represented by optical techniques such as video-extensometers or Digital Imaging Correlation (DIC), which require some markers on the surface of the sample as reference to measure the displacements.
- **Grips**: Pneumatic, mechanic or externally controlled hydraulic clamps that provide the mechanical connection between the specimen and the tensile testing machine. Their function is to transfer the movement of the cross head to the specimen and transfer the test force generated in the specimen to the load cell.
  In Dallara, hydraulic wedge grips are used to guarantee a stronger grip, suitable for high performance materials, such as composites.
- **Software**: It controls the machine allowing the user to set up the testing system, configure and run tests, store and display the results. The amount of force applied to the machine and the elongation of the specimen are recorded throughout the test.

3.3 Standard

3.3.1 Tensile Test

The tensile test is performed according to the ASTM D3039 standard which defines everything that is needed in order to perform the test correctly and in a way so that to obtain results easily comparable. In particular the standard states that:

> A thin flat strip of material having a constant rectangular cross section is mounted in the grips of a mechanical testing machine and monotonically loaded in tension while recording the force. The ultimate strength of the material can be determined from the maximum force carried before failure. If the coupon strain is monitored with strain or displacement transducers then the stress-strain response of the material can be determined, from which the ultimate tensile strain, tensile modulus of elasticity, Poisson's ratio, and transition strain can be derived.

The standard prescribes to test at least five specimens per test condition unless valid results can be gained through the use of fewer specimens, such as in the case of a designed experiment. However, it does not provide a specific geometry for the samples because of lack of industry consensus, but some general requirements, in order to obtain correct result, exist. In particular

width and thickness should be selected in order to promote failure in the gauge section and ensure a significant amount of material under stress in the gauge length is present to produce a more statistically significant result. However, specific geometries, described below, have been found by a number of testing laboratories to produce acceptable failure modes.

Fiber Orientation	Width, mm [in.]	Overall Length, mm [in.]	Thickness, mm [in.]	Tab Length, mm [in.]	Tab Thickness, mm [in.]	Tab Bevel Angle,°
0° unidirectional	15 [0.5]	250 [10.0]	1.0 [0.040]	56 [2.25]	1.5 [0.062]	7 or 90
90° unidirectional	25 [1.0]	175 [7.0]	2.0 [0.080]	25 [1.0]	1.5 [0.062]	90
balanced and symmetric	25 [1.0]	250 [10.0]	2.5 [0.100]	emery cloth	—	—
random-discontinuous	25 [1.0]	250 [10.0]	2.5 [0.100]	emery cloth	—	—

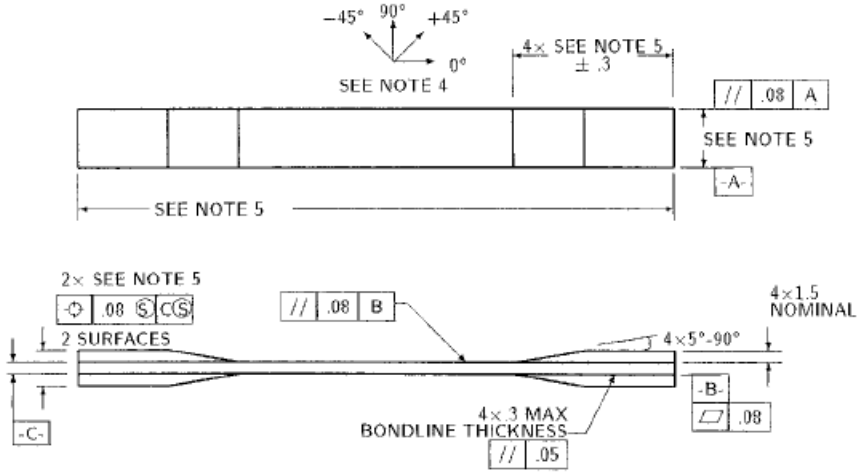


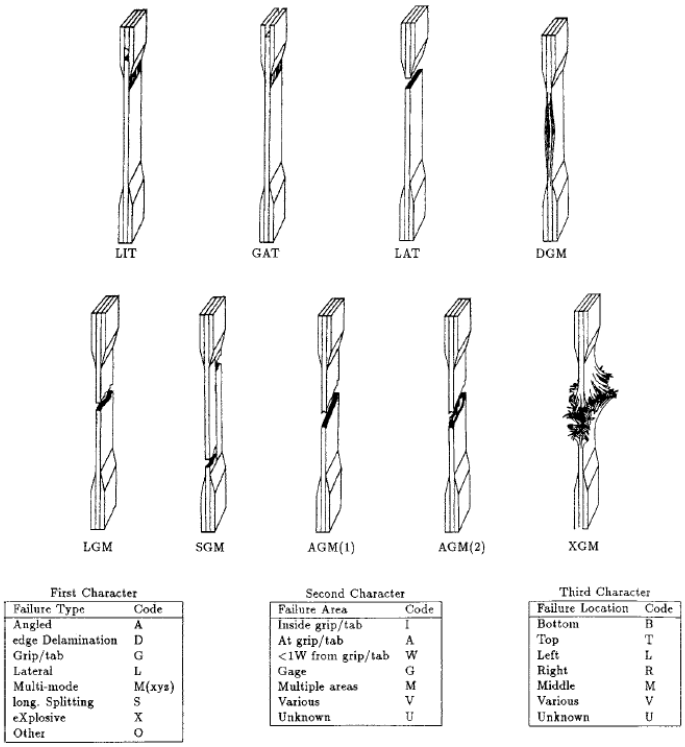
Figure 16: Recommended sample geometries for a tensile coupon

In order to characterize the elastic properties of the laminate the coupon is realized by unidirectional fibers oriented in different directions. In particular, to derive the longitudinal and transversal properties, 0° and 90° oriented fibers are respectively employed. Whereas, to obtain shear properties, a 45° coupon layup is used according to the standard ASTM D3518 , which however recall the same speed and geometry of the present standard.

Regarding tabs, these are strongly recommended when testing unidirectional materials, whose geometry is again reported in the table above (26). The tab are realized in a carbon $[0/90]_{ns}$ laminate configuration and then applied at 45° to the force direction to provide a soft interface. In our case, because of the manufacturing of the coupons, only the tab adhesive was tapered.

The testing speed should be set in order to induce a nearly constant strain rate in the gauge section. Specifically, the strain rate should be selected so as to produce failure within 1 to 10 min: in particular for the suggested geometry, the strain-controlled tests should be performed with a strain rate of 0.01 min^{-1}.
Beside the geometry and the speed of testing, also the accepted type of failures are described. In particular the failure mechanisms are classified through a three part failure mode code. It consist in 3 letters (fig. 17:

1. the first define the type of failure
2. the second define the area of the failure
3. the third define the location (top, bottom, middle...)



First Character

Failure Type	Code
Angled	A
edge Delamination	D
Grip/tab	G
Lateral	L
Multi-mode	M(xyz)
long. Splitting	S
eXplosive	X
Other	O

Second Character

Failure Area	Code
Inside grip/tab	I
At grip/tab	A
<1W from grip/tab	W
Gage	G
Multiple areas	M
Various	V
Unknown	U

Third Character

Failure Location	Code
Bottom	B
Top	T
Left	L
Right	R
Middle	M
Various	V
Unknown	U

Figure 17: Three part code of acceptable failures.

It is important to note that, since the main goal of the test is to obtain a failure in the free length of the sample, an excessive failure in the tab/grip regions is not accepted.

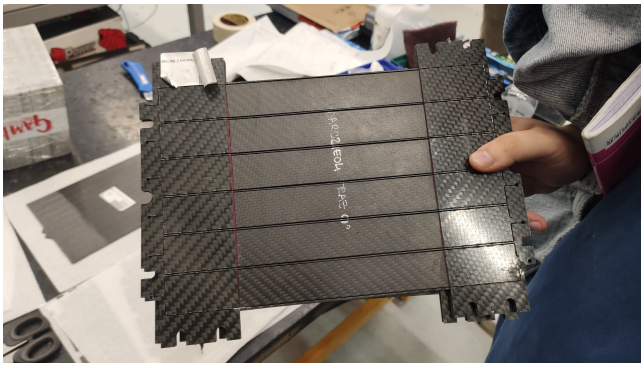


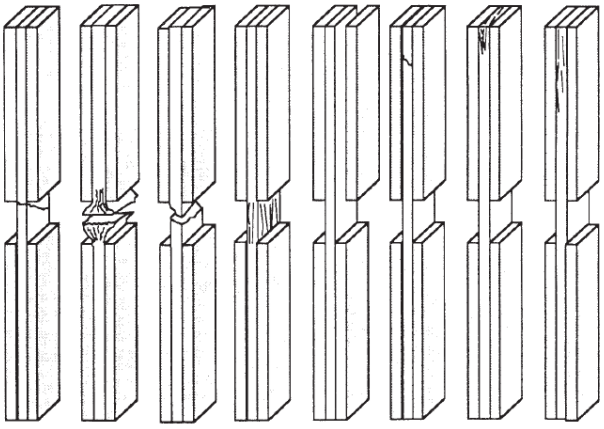
Figure 18: Specimens with tabs bonded using adhesive

3.3.2 Compression Test

Compression tests can be performed following different standards: ASTM D3410, ASTM D6641 or SACMA 1R-94.

ASTM D3410 Specific for the in-plane compressive properties of polymer matrix composite materials reinforced by high modulus fibers, the defined test procedure introduces the compressive load through shear at wedge grip interfaces using specially designed fixtures. Indeed rectangular wedges, allowing for variable width and thickness specimens, are used and sometimes present slots at the outer ends to accommodate end bars. The ends of the specimen can then be positioned in contact with these bars during grip screw tightening, ensuring equal

lengths of the specimen gripped on each end. Moreover, if these bars are not removed prior to testing, they will provide a certain amount of end loading, together with the previously mentioned shear one. These bars also promote equal movement of each of the wedges of a pair during specimen loading, reducing induced bending.
Finally, because such test is sensitive to the alignment of the sample with respect to the wedges, an alignment jig may be employed to mechanically hold it outside the fixture housing blocks. A flat strip of material with constant rectangular section is used as sample, which is inserted between the grips and loaded in compression until failure, allowing to measure the ultimate compressive strength.
Finally, following a similar approach as the one described for the tensile standard, a Three-Part Failure Identification Code procedure is defined to record the mode, area and location of failure on the specimen. Again, the most desirable failure area is the gauge section, since gripping and tabbing influences are minimal there. However, considering the small gauge length for such test, failure in the gauge section near the tabs is common and still acceptable.



TAT BGM HAT SGV
Acceptable Failure Modes and Areas

DTT HIT CIT DIT
Unacceptable Failure Modes and Areas

First Character

Failure Mode	Code
Angled	A
Brooming	B
end-Crushing	C
Delamination	D
Euler buckling	E
tHrough-thickness	H
Kink bands	K
Lateral	L
Multi-mode	M(xyz)
long.-Splitting	S
Transverse shear	T
eXplosive	X
Other	O

Second Character

Failure Area	Code
Inside grip/tab	I
At grip/tab	A
Gage	G
Multiple Areas	M
Tab adhesive	T
Various	V
Unknown	U

Third Character

Failure Location	Code
Bottom	B
Top	T
Left	L
Right	R
Middle	M
Various	V
Unknown	U

Figure 19: Compression specimen failure identification codes and overall specimen failure schematics

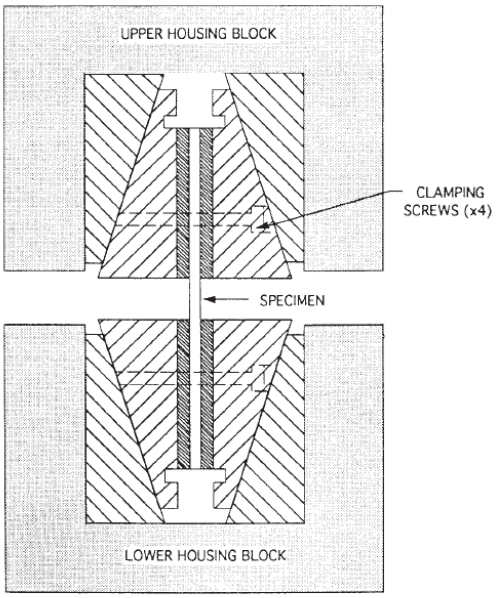


FIG. 5 Schematic of Compression Test Fixture

Figure 20: Compression test specimen with tabs, which are however not required unless acceptable failure modes and locations are not recorded. Particular attention should be paid to parallelism and flatness of the tabs, often requiring precision grinding.

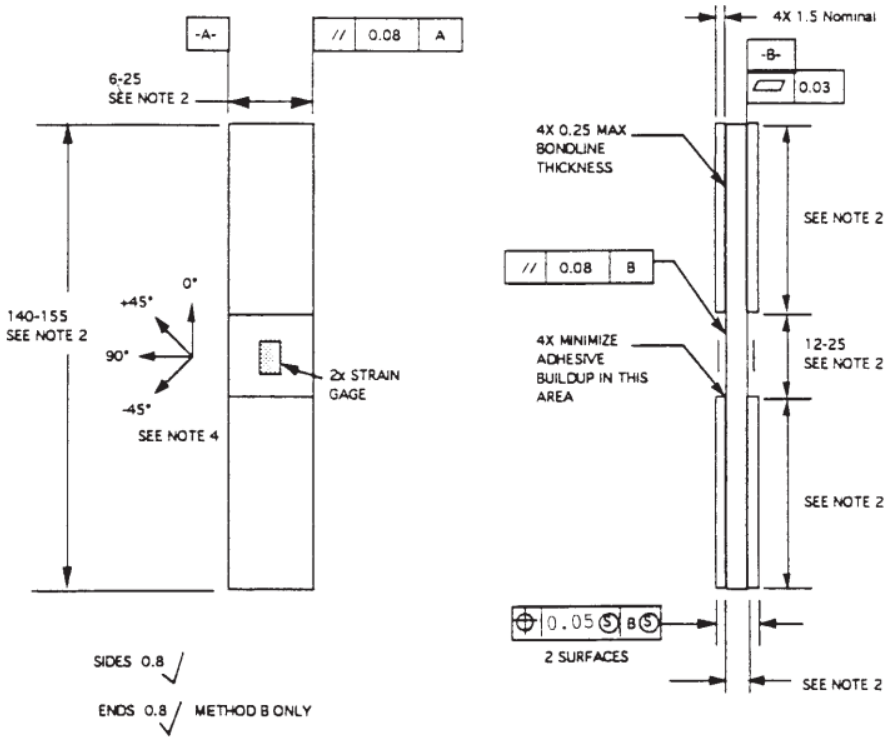


Figure 21: Compression test specimen with tabs, which are however not required unless acceptable failure modes and locations are not recorded. Particular attention should be paid to parallelism and flatness of the tabs, often requiring precision grinding.

SACMA 1R-94 Modification by Boeing of the already exiting ASTM D695, this standard applies primarily to prepregs with the target of obtaining compressive strength or modulus,

based on the sample employed, through end loading.
This is achieved using a specific fixture including two parallel "V" grooved plates supporting the specimen and hindering the risk of buckling due to end loading on a vertical plane. Moreover grooves are used to reduce fractional loading on the sample when the plates are being compressed together.

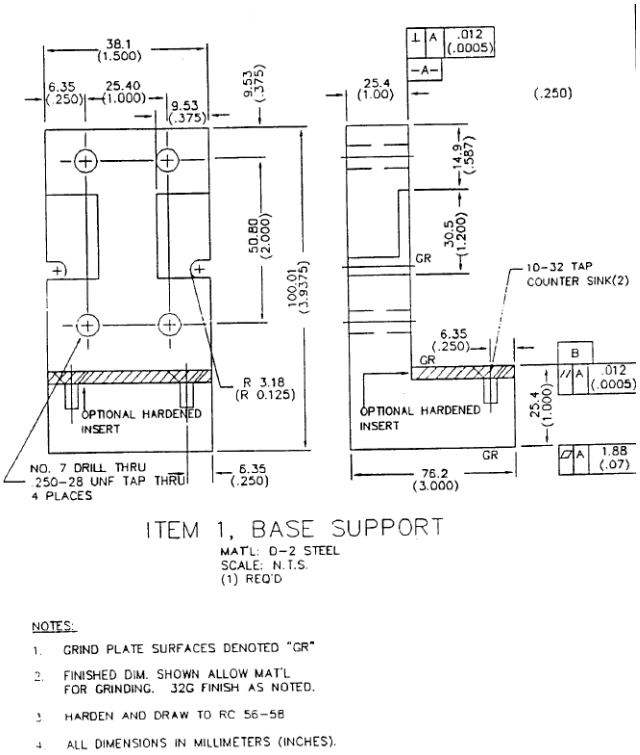


Figure 22: Compressive modulus test specimen.

The specimen will thus be placed between the platens of the testing machine and loaded in compression until failure, when the scope is to obtain the ultimate compressive strength, otherwise the load-strain curves may be retrieved during a separate test if the modulus is desired.
At least five specimen should be tested per test condition, with a load of 1 mm/min until 5000 microstrain for modulus measurement, otherwise until failure. Results shall not be considered for any specimen which fails due to flaws such as tab debonds or failure within the tab area. In such tests, particular attention should be paid to fiber alignment, surface quality and, because of the end loading definition, parallelism of the specimen ends as well as perpendicularity and system alignment.

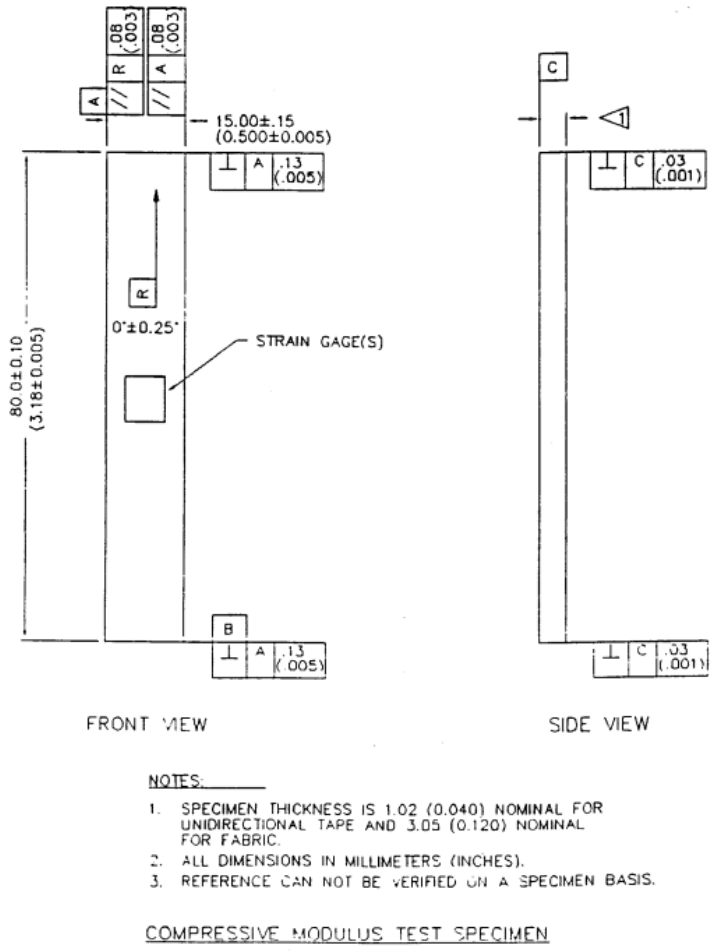


Figure 23: Compressive modulus test specimen.

ASTM D6641 This standard is applicable to flat laminates with balanced and symmetric layups with at least a single 0° ply, no tabs are required. With respect to the ASTM D3410 (pure shear loading compression test) and ASTM 695 (pure end loading), here the specimen is subjected to compression through a combination of both shear and end loading. This is achieved using a special fixture which is itself loaded in compression between the flat grips of a universal testing machine. It also allows to control the ratio between end and shear loading through the torque applied to the fixing screws.
The load shall be applied at a nominal rate of 1.3 mm/min until failure which should happen between 1 to 10 min, otherwise if only the modulus is being determined, the load should be applied approximately 10 % beyond the upper end of the used strain range.
At least five specimens per test condition need to be tested unless fewer specimens allow for valid results.

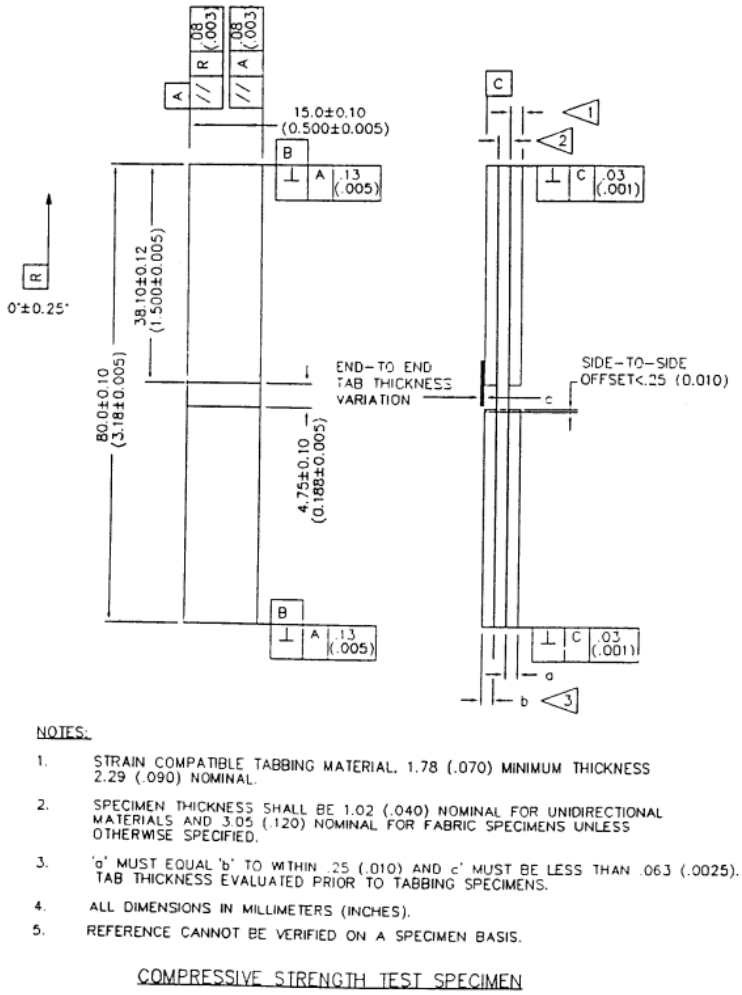


Figure 24: Compressive strength test specimen. Tabs shall be in composite material with [0/90] orientation.

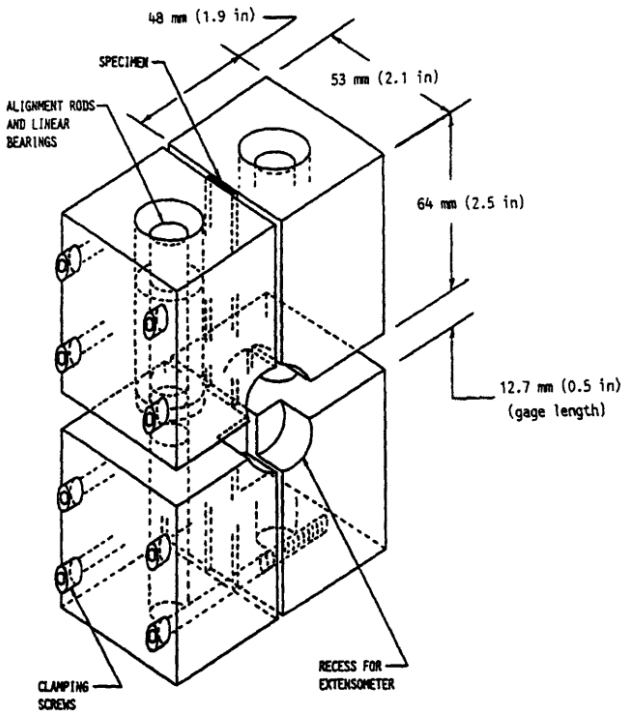


Figure 25: Typical Combined Loading compression test fixture

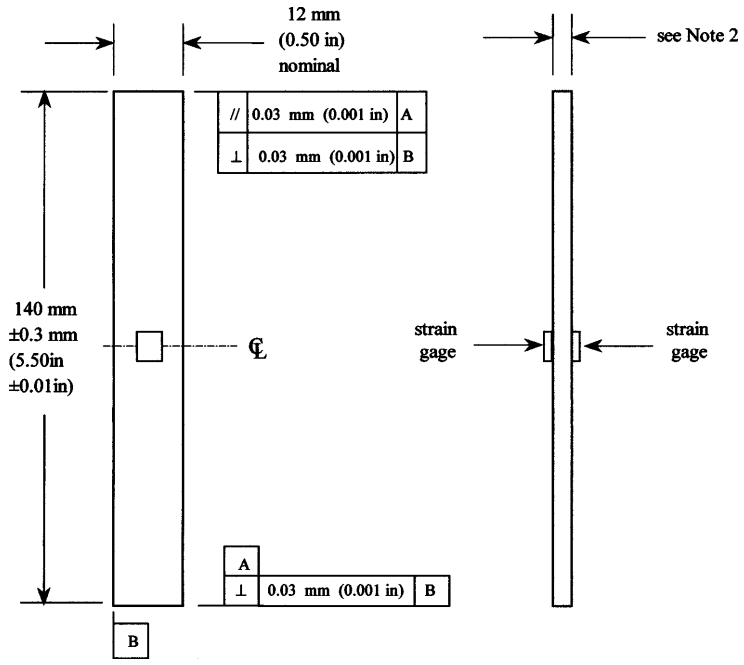


Figure 26: Typical test specimen configuration. Different width may be employed, usually higher than the 12 mm nominal value, to maintain a representative gauge section volume of material being tested. No specific thickness is defined, however it should be uniform and enough to avoid Euler column buckling. Finally, faces can be slightly lapped to remove local imperfections

3.4 Laboratory activity

First of all, few words should be spent on the manufacturing of the coupons. In order to guarantee a consistent alignment of all the samples fibers and of the tabs, to minimize possible errors of positioning between one coupon and the other, the preparation of the samples start from a monolithic tabbed laminate. Therefore, starting from a laminate sheet, the tabs are bonded using a machine that applies heat and compression to allow proper bonding of the adhesive. Once this is done, the individual samples are obtained cutting the laminate using a water jet (Fig. 18).

During the experimental characterization, two main kind of composites systems (fiber + matrix) were elected to be investigated:

- T300 ER434: T300 fiber has small tows that typically go from 1k to 3k, with 2/2 twill, designating it as a very tight material. The matrix ER434 is a specific resin for impacts, used in absorbing structures
- T800 ER450: T800 fiber on the other hand has very large tows, usually 24k, meaning that it is more flat with respect to the other enhancing in the end the mechanical properties of the material, with 2/2 twill. ER450 is a tougher matrix with respect to the previous one and it is characterized by good thermal properties (it can be cured from low temperatures) with the disadvantage, however, of having a lower fracture toughness.

In particular, 18 samples for each material were tested arranged in batches of 6 coupons for each orientation (0°,90°,45°). The nominal dimensions used for this tensile test are reported in the table 5 below.

free length	150
overall length	250
width	25
thickness	2

Table 5: Nominal dimensions of the coupons

Before performing the characterization of the samples, however, it is necessary to measure their dimensions, irrespectively from the nominal ones. Indeed this information will be important for the extrapolation of the mechanical properties of the material. An average of three readings each of width, thickness and free length is recommended. The opportunity of measuring can also be taken to examine the quality of the specimen: those with obvious notches or other machining damage should be discarded.

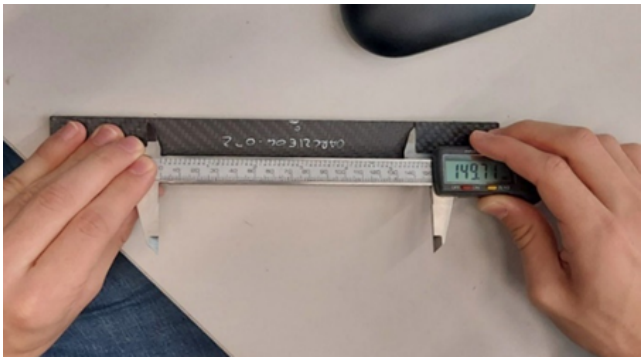


Figure 27: Specimens measurements using a caliber

3.4.1 Tensile Test

Then the samples (one at time, for each test) are mounted on the testing machine. They are fixed through hydraulic grips, because of the mechanical or pneumatic ones limit on the maximum loads that can be carried out by the grips itself, in presence of such a high performance materials. The pressure of the grips can be controlled by a manifold depending on the material and the expecting maximum load, fundamental to apply the correct pressure:

- a too low pressure will provide an insufficient force that will permit the specimen to slip, a behaviour which can be detected in the stress strain curve by the presence of a plateau.
- a too high pressure instead provide an excessive clamping force risk to induce a distortion of the end-tab outer fibers, leading to a reduction of the effective failure load.

During the mounting process of the samples, particular attention must be paid on the correct positioning with respect to the axis of the testing machine. Indeed, even small, may result in large local stresses and bending effects, so that to affect negatively the reliability of the results. So to avoid such problems, we should ensure not only the relative alignment of the grips but also the alignment of the sample with respect to them. Alignment of the grips relative to each other is determined by the machine on which they are mounted, since manufacturer adopt very rigid clamps in a stiff testing frame thus to ensure that alignment is 'built in' to the machine. The the centreline of the specimen should be manually aligned with the axis of the testing machine so as to eliminate bending and asymmetric loading.

Once mounted, it is possible to mount the extensometer on the free length of the specimen. Before starting the test the initial gauge length should be measured.

In particular, in the activity performed, the samples were inserted into the machine by controlling manually the height of the crosshead and then tensioning it with a value of the load cell around 50-100 N in the ready-to-test condition to provide a correct data acquisition. in the meanwhile the sample was clamped on its tabs with a pressure of about 80 bar (a quite high pressure, but justified by the expected high mechanical properties).

Before starting the test, the software that control the machine and evaluate the signals coming from the sensor of the testing machine must be set. It requires:

- the geometric data of the sample: in this way the software elaborate the signals coming from the load cell and the LVDT transducer in terms of stresses and strains;
- gauge length of the extensometer: in this way it's possible to evaluate the sample displacement removing the cross head displacement;
- the loading rate in terms of the testing machine grips (grip separation rate): 2 mm/min
- maximum and minimum load of the load cell: in this way the test is stopped if a value of maximum (or minimum) load is reached to preserve the machine or if the failure of the sample is reached.

The failure is evaluated by observing the difference between the step load value n and n-1, if this delta is larger than 90% the test is stopped.
The extensometer must be removed from the sample before reaching the failure load of the

specimen in order to avoid to damage this expensive tool as a consequence of the explosive failure that characterize this kind of test. basing on the experience the extensometer is removed when a load of 10-15 kN is reached, Therefore, in phase of post processing, a correction will be performed when evaluating the Young's modulus.
At the beginning of the test the software put to 0 all the parameters except for the load because it keeps the information related to the preload of the sample.

Once performed the test (for each sample), the specimen must be analysed qualitatively in order to observe where the failure occurred, to define if it is acceptable, according to the standard: failure in the free length and near the tab ends are accepted while for failures in the tab region, the coupon and its results must be discarded.

3.4.2 Compression Test

As for the tensile test, the electromechanical MTS machine was employed, together with MTS 643 compression plates, allowing for tight parallelism tolerances between upper and lower plates. SACMA 1R-94 fixtures were used and, following the standard procedure described above, the samples are placed into the anti-buckling structures while paying particular attention to the whole system alignment to avoid influence due to bending.
Finally, the specifics of the test are inserted in the controlling software:

- Test speed
- Sampling frequency [HZ]
- Deformation limit, set to 30% for safety reasons
- Load application at 1 mm/min
- Sample specifics, test type and costumer

Again, once the test have been completed, a first qualitative inspection of the broken samples was made, which confirmed mostly acceptable "V" shaped compressive failures for both composites batches. However, a few under tab failure were detected and deemed unacceptable.

3.5 Post processing

Once the tests have been performed and the data acquired, the latter must be elaborated in order to extrapolate the mechanical properties of the material that will be used in the analytical and numerical model to asses the design of the components.

From the testing machine 5 kind of data are acquired from the respective channels:

- Cross head displacement [mm] (ε_{CH})
- Extensometer displacement [mm] (ε_{ext})
- Load [kN] (P)
- Transversal displacement [mm] (ε_{e_2})
- Time [s]

which are used along with the input geometric dimension to calculate the properties of the material.
In order to elaborate the raw data provided by the signal acquisition some calculations and consideration must be carried out. In particular, since the LVDT sensor will return the global displacement of cross-head + coupon, in order to extrapolate the effective deformation experienced by the material the compliance of the machine must be taken into account.

The cross head strain is the strain computed considering just the stroke of the testing machine, which is affected by an error due to the compliance of the machine that is not negligible.

$$\varepsilon_{CH} = \frac{cross\ head\ displacement}{free\ length}$$

The extensometer strain is computed considering just the displacement measured by the extensometer (could be both the biax. ext. or the video ext.).

$$\varepsilon_{CH} = \frac{extensometer\ displacement}{gauge\ length}$$

The stress σ is the stress measured at the sample cross section.

$$\varepsilon_{CH} = \frac{Load}{Cross\ secional\ area}$$

It can be considered as the actual stress level since the load measured by the load cell is not affected by the compliance of the testing machine. The only error in the computation of the stress can be obtained by a non-accurate mounting of the coupon on the testing machine (e.g. a small deviation from the straight position of the coupon can generate moments on the sample leading to a measured total load no more coherent with the axial stress).

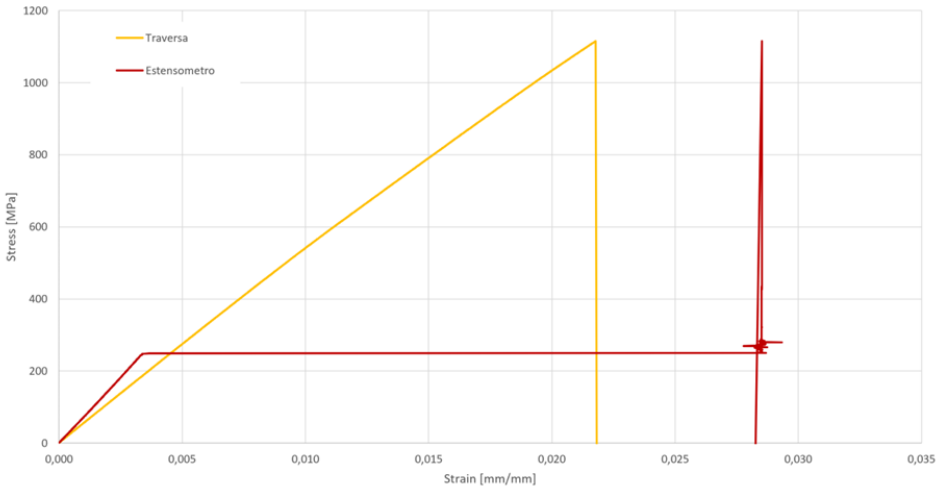


Figure 28

The yellow one is a $\sigma - \varepsilon_{CH}$ curve, while the red is a $\sigma - \varepsilon_{ext}$. In practical terms, the first is the one measured by the LVDT while the second one is measured by the extensometer, where the step of the red curve is due to the dismounting of the biax. extensometer from the sample. There is an evident difference between the red and yellow slope, thus leading to a difference in the young modulus computed. This difference is simply due to the compliance of the testing machine. So we need a correction. The values of the two elastic moduli E_{CH} and E_{ext} are

computed through a linear regression of two points taken as reference, inside the the linear trend of their respective $\sigma - \varepsilon_i$ curves.

Then in order to compute the displacement of experienced by the sample, the contribution of the compliant cross head must be removed. To do that, first of all, the the difference in displacement between cross head (the overall one) and the extensometer is calculate, thus returning the the strain of the cross-head alone, that will be useful to calculate the stiffness of the testing machine:

$$\varepsilon_{extra\ error} = \varepsilon_{CH} - \varepsilon_{ext}$$

Secondly, the elasticity K_{CH} of the cross-head is computed calculating the slope of the linear part of the $F - \varepsilon_{extraerror}$ curve between the same reference points, hence making possible to compute the effective cross-head contribution to the displacement from which finally we can yield the corrected strain of the sample:

$$\varepsilon_{computed} = \frac{laod}{K_{CH}}$$

$$\varepsilon_1 = \varepsilon_{CH} - \varepsilon_{computed}$$

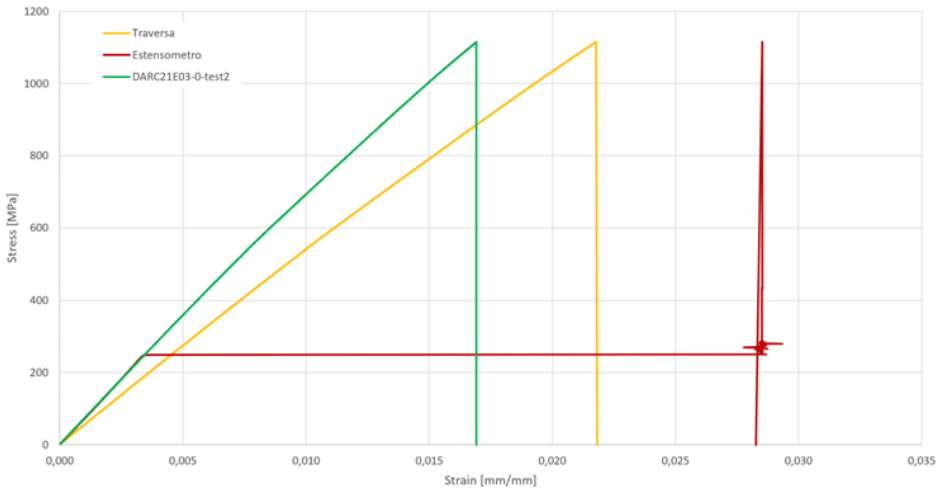


Figure 29

The green curve, the correct stress-strain curve experienced by the sample, is finally obtained, by simply subtracting the strain, due to the elasticity of the testing machine, to the yellow one (the global strain). It shows a different slope from the one of measured by the LVDT transducer and above all a completely different elongation at break, for the same σ_{max}.

From the definitive Strain-Stress Curve it is possible to compute the mechanical properties of the sample that will composed the material card of the material under investigation.

- E_x (the correct elastic modulus of the material tested; it can be equal to E_1 or E_2 according to the 0° or 90° orientation of the fibers): it is obtained as linear regression between two reference points (σ,ε) of a $\sigma - \varepsilon_1$ curve.

- **Poisson ratio** ν_{xy} (equal to ν_{12} or ν_{21} according to a 0° or 90° laminate): it is computed as the ratio between the transverse strain $\varepsilon_t = \frac{transverse\ displacement}{width}$ and the longitudinal one returned by the extensometer:
$$\nu = \frac{\varepsilon_t}{\varepsilon_e xt}$$
an averaged value is then extrapolated basing on its values in correspondence of the reference values on the $\sigma - \varepsilon_1$ curve.
- **failure stress** σ_u, computed as the maximum value among the stresses σ
- ε_f is the strain associated to σ_f

ε rottura [mm/mm]	0,011
σ rottura [MPa]	551,4
E corretto [MPa]	52721
v [mm/mm]	0,000
σ snervamento [MPa]	551,4

Figure 30: example of extrapolated mechanical properties of a sample

3.6 Results analysis

Finally the results elaborated can be compared. In particular, since we are testing 6 specimens per test (fiber orientation and material) a variability of the results due to the geometry variation, material and manufacturing affecting factors are expected. For this reason the properties of the material, coming from the coupon results, are analysed in terms of average values, standard deviation and coefficient of variation. The average values will be used to build up the material card, while the standard deviation and the coefficient of variation (that gives an information about the repeatability of the behaviour of a property: the lower the C.O.V the more consistent is the result) provide information about the dispersion and reliability of the test results.

3.6.1 Tensile Test

Looking at the fibers datasheets, shown in table 6, it's possible to see that T800 shows higher mechanical properties in terms of strength and modulus:

	T300	**T800**
Tensile Strength [Mpa]	3530	5490
Tensile modulus [Gpa]	230	294

Table 6: fibers datasheet data

Since we're testing fabrics with same waving (twill 2x2) we expect the same behaviour of the properties also in the tensile test.

Comparing the results of the two materials in the same direction we can observe

- **0° test:** The stress at brake of T800 is more than double with respect the T300 reaching respectively 1176 MPa and 560 MPa that are reasonable and good results for these two fibers.

	T300			T800		
	Avg	Std Dev	COV %	Avg	Std Dev	COV %
εR [mm/mm]	0,008	0,001	10,955	0,018	0,001	4,328
σR [MPa]	560,3	28,4	4,606	1176,3	36,6	3,112
E [MPa]	54161	1158	2,043	73689	1790	2,430

Figure 31

- **90° test:** same consideration can be done for the comparison between these two materials along 90°.

	T300			T800		
	Avg	Std Dev	COV %	Avg	Std Dev	COV %
εR [mm/mm]	0,012	0,001	10,955	0,020	6,321	31757,185
σR [MPa]	615,7	28,4	4,606	1264,2	346,2	27,381
E [MPa]	56701	1158	2,043	72176	32407	44,900

Figure 32

- **45° test:** T800 is impregnated with ER450 that is a classical epoxy resin used for structural component, while T300 is impregnated with ER434, that is a toughened epoxy resin useful for impact and energy absorption. In this case the elongation at brake of T300-ER434 is higher than T800-ER450 as well as the sigma at failure, leading to an higher plastic region. This behaviour can be interpreted as an higher amount of energy absorbed and it is consistent with the nature of the resins.

	T300			T800		
	Avg	Std Dev	COV %	Avg	Std Dev	COV %
εR [mm/mm]	0,120	0,006	5,059	0,074	0,005	6,455
τR_45° [MPa]	121,3	2,0	1,618	90,4	1,1	1,216
G [MPa]	3502	77	2,197	210	0	0,058

Figure 33

Last but not least, it is interesting to note that the properties at 90° are slightly higher respect to the ones at 0°. The difference can be probably due to the manufacturing process of the fabric in the realization of the warp and weft.

3.6.2 Compression Test

Firstly, the T300 ER434 samples were analysed for both fibre orientations.
Considering the excessive number of unacceptable under tab failures for the 0° batch, 8 out of

14 total samples, it is reasonable to assume the presence of issues in the tabs bonding.
On the contrary, the 90° set shows a much more consistent behaviour between all samples, with a single unacceptable failure.

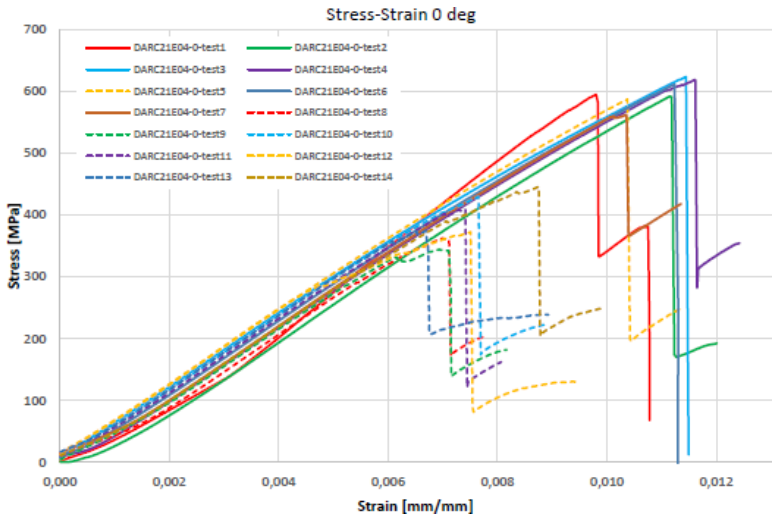


Figure 34: Stress-Strain curves - T300 ER434 0°

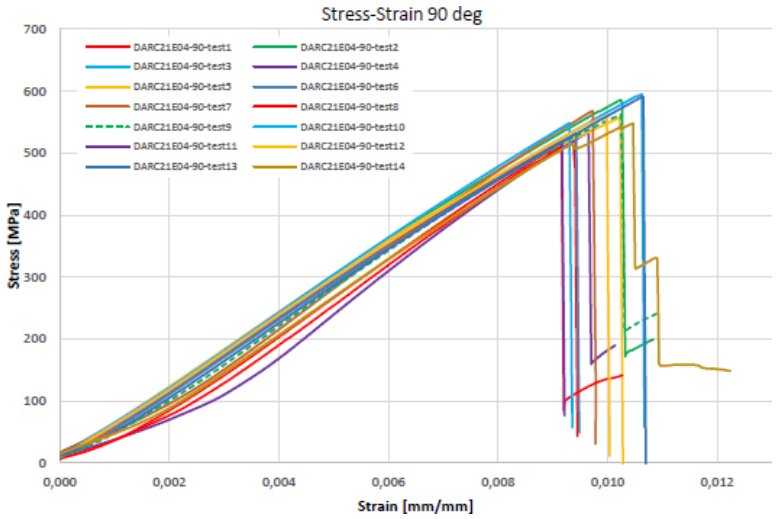
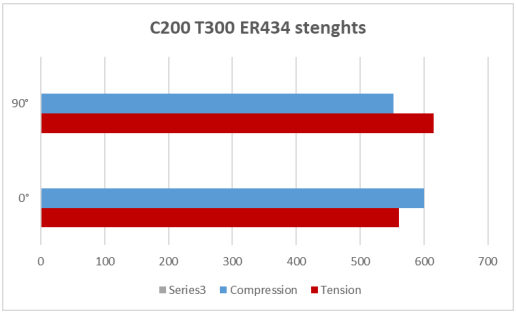


Figure 35: Stress-Strain curves - T300 ER434 90°

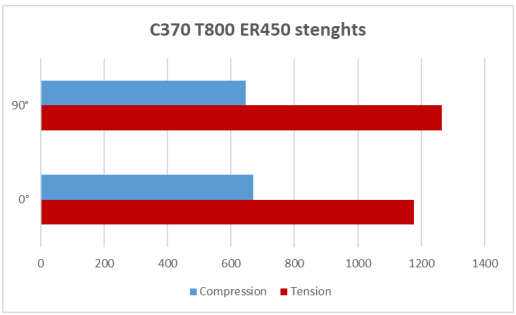
Focusing then on the T800 ER450, 6 failures were again deemed unacceptable for the 0° set, while only 4 for the 90°.

It is interesting to notice how the fiber orientation influence on the strength is the opposite with respect to tensile tests, with 0° batches showing slightly improved characteristics. Finally, for the T300 ER434, in the 0° direction, a higher compressive strength is measured with respect to the tensile one. Comparing instead the compression behaviour of the two different coupons it is possible to note that the compressive strength of the T800 is generally higher becuase of the tougher matrix used.



	T300	0°	90°	45°
Tension	εR [mm/mm]	0,008	0,012	0,120
	σR [MPa]	560,29	615,69	121,33
	E [MPa]	54161	56701	3502
Compression	εR [mm/mm]	0,01	0,01	/
	σR [MPa]	599,96	552,45	/

Figure 36: Stress-Strain curves - T300 ER434 0° vs 90°



	T800	0°	90°	45°
Tension	εR [mm/mm]	0,018	0,020	0,074
	σR [MPa]	1176,34	1264,21	90,41
	E [MPa]	73689	72176	210
Compression	εR [mm/mm]	0,01	0,01	/
	σR [MPa]	670,34	647,13	/

Figure 37: Stress-Strain curves - T800 ER450 0° vs 90°

4 Single Lap Joint Shear Test

4.1 Introduction

Single Lap Shear joint Test is a destructive technique used to evaluate the strength of an adhesive joint. In particular pulling the two substrates bonded in a single lap configuration it is possible to evaluate how the joint behaves for different overlap lengths, thicknesses and surface treatments of the adherents. In this way not only the strength but also the different affecting factors can be taken into account.

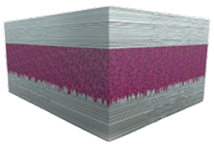
4.2 Theory

An adhesive is a non metallic material capable of joining bodies through a surface contact (adhesion) and an intrinsic bond (cohesion) without changing the essential structure of the bodies. Therefore the bonding strength is strongly dependent on these two principles:

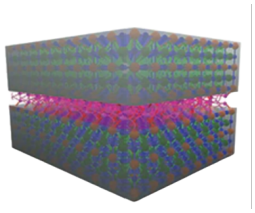
- **Adhesion**: Interlocking of adhesive and substrate at the interface
- **Cohesion**: Adhesive's ability to hold itself together under stress (adhesive molecular interaction)

In particular adhesion can be characterized by several mechanisms:

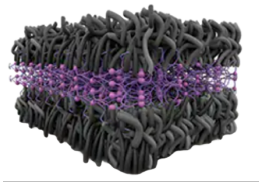
Mechanical interlocking: Occurring when the adhesive flows into the pores of the substrate surface. The interlocking provides a physical impediment to crack propagation at the interface.



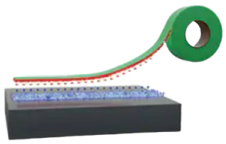
Chemical adhesion: It's the strongest mechanism of adhesion, with bonds occurring between atoms/molecules on the substrate surface. This mechanism contributes to the structural strength of many adhesives such as epoxies bonding aluminium.



Diffusion adhesion: It takes place when the adhesive polymer is capable to penetrate and lock with a polymeric substrate. An interface is created with entangled polymer chains bonding adhesive and substrate. Commonly used in difficult to bond low surface energy materials such as polypropylene.



Electrostatic adhesion: It occurs as the charged surface of the adhesive is drawn toward the oppositely charged substrate, most common in tape applications.



Based on the working principles of adhesives it is possible to derive some useful tips allowing to increase the quality of a bonded joint, acting on its constituents. In particular it depends on 3 basic guidelines:

- **surface preparation:** in order to improve the quality of the bonding the surface must be treated chemically and/or physically, by cleaning it with suitable chemical agents to remove the dirt which acts as local defect and by increasing the surface roughness to provide the interlocking points and a wider area of adhesion;
- **wettability:** it is a material dependent property which refers to the surface energy of the material and it is identified as the capability of a liquid to spread uniformly over the material surface. the higher the wettability, the higher the quality of the bonding (epoxies, CFRPs and steel and Al alloys are characterized by an high wettability);
- **Pressure application:** in order to yield a good penetration of the adhesive on the roughed surface and let the air, trapped in between the adhesive layer and the substrate, getting out;

However, beside the material and surface properties of the joint, it is possible to increase the strength of the adhesive bond acting also on the geometry. This constitutes a very critical aspects to take into account since a wrong geometry can weak the bonding and induce unfavorable stresses which tend to separate the substrates (peel and cleavage stresses). For those reasons, two main rule of thumbs must be followed during the design of a bonded joint:

- to maximize the bonded area
- to minimize the peel and cleavage stresses, while exploiting compressive stresses.

From a mathematical point of view, the single lap joint is nominally loaded in shear, but actually, because of the eccentricity due to the geometry, also a bending moment is experienced. This generate the typical peel stresses that tend to separate the substrates from the adhesive.

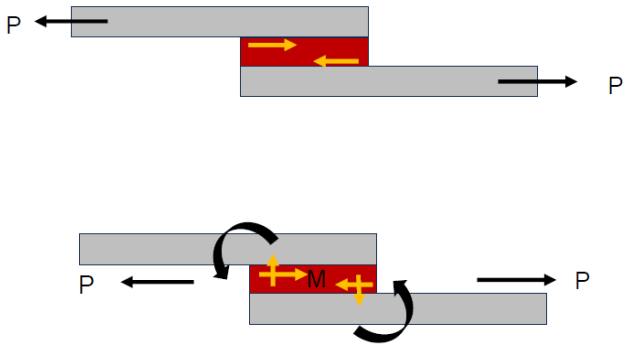


Figure 38: Loading geometry

Both the shear and the peel stresses are affected by the material and the geometry of the joint. Indeed, defining a stress concentration factor as the ratio between the maximum component of stress under scrutiny and its average, it's possible to predict how the behaviour of the joint will change with the geometric parameters and material properties. However, since in this laboratory activity only the thickness of the joint is changed, only its effect is taken into account.

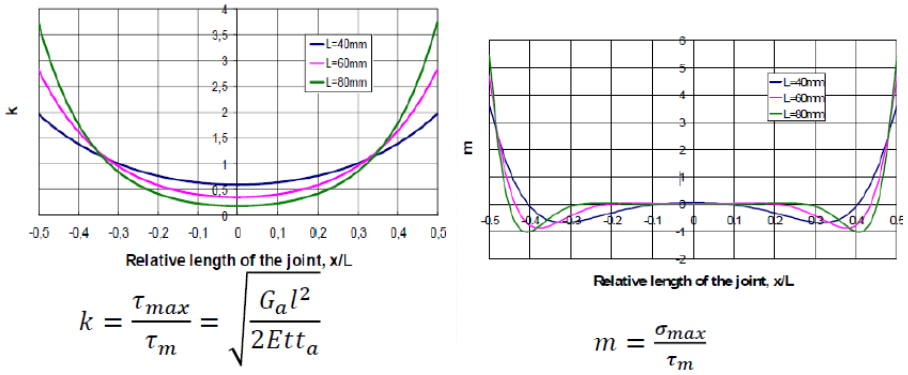


Figure 39: stress concentration distribution on the overlap length

$$m = \frac{\sigma_{max} * k}{\tau_{max}}$$

In particular, it is possible to see that, for the shear stresses, the thickness has a beneficial effect while for the peel ones, an increase of thickness lead to an increase of the eccentricity of the geometry, thus emphasizing the concentration of this kind of stresses. From the graph of fig.39 it is possible also to observe that the most stressed area is located at the edges of the bonding, where will be expected to obtain the failure.

According to the overall conditions we can experience different type of joint failure:

- **adhesive failure - adhesive strength $>$ interlaminar forces:** The adhesive is detached from the adherent, without leaving any trace on it. This means that the forces of adhesion are lower than those of cohesion.

- **cohesive failure: adhesive strength $<$ interface forces:** The rupture occurs inside the adhesive. This is usually the condition where the maximum joint strength is achieved. In this case we exploit all the strength of the adhesive until it breaks.

- **mixed mode failure:** This type of failure occurs when the forces of adhesion and cohesion are similar and small "local" differences determine the path of rupture
- **substrate delamination: substrate strength $<$ adhesive joint strength:** typical of composite adherents, this occurs when both the adhesive and cohesive forces are far stronger than the inner strength of the material attached.

4.3 Activity objective

The aim of the testing activity was to evaluate the behavior of an adhesive joint in which two aluminum plates (Al 7075 T6) was bonded together through a structural epoxy adhesive (EC-9323), in a single lap joint configuration. In particular the test was performed for different adhesive thicknesses in order to evaluate the strength dependency on this parameter.

The adhesive consist in a two-component epoxy paste adhesive which cure at room temperature or with mild heat to form a tough, impact resistant structural bond. It has an excellent adhesion to a wide variety of substrates such as metals, glass, ceramics and plastics, incl. GFRP and CFRP. Once cured it provides extremely high shear and peel strength over a wide temperature range.

A reference strength value for the joint was taken from a datasheet provided by 3M where the same adhesive was used together with 2024 T3 aluminium plates, where the sample have been cured for 15 days at room temperature. The results reported was based on a test performed according to the standard EN 2243-1:

Overlap shear strength	Test temperature	Results	Test method
Al 2043 T3	23°C	36 MPa	EN 2243-1

Table 7: 3M datasheet

4.4 Standard and specimens

The single lap joint test was performed according to the ASTM D1002 standard which is comparative in nature, and not correlated to real world performance, since different environmental conditions and joint geometries contribute to the variation of adhesive performance. It is a testing method developed to measure the shear strength of adhesives that are used to bond metals. This test is similar to ASTM D3163 which is for adhesives that bond rigid plastic substrates, and also ASTM D3164 which is for plastic adhesives that are used to bond both plastic and metal substrates. All three of these specifications use a single lap joint (lap shear) specimen to determine the shear strength of adhesives.

The recommended thickness of the sheets is 1.62, even if Dallara used plates of 3 mm for historical reasons. The recommended length of overlap for most metals is 12.7 $\pm$ 0.25 mm.

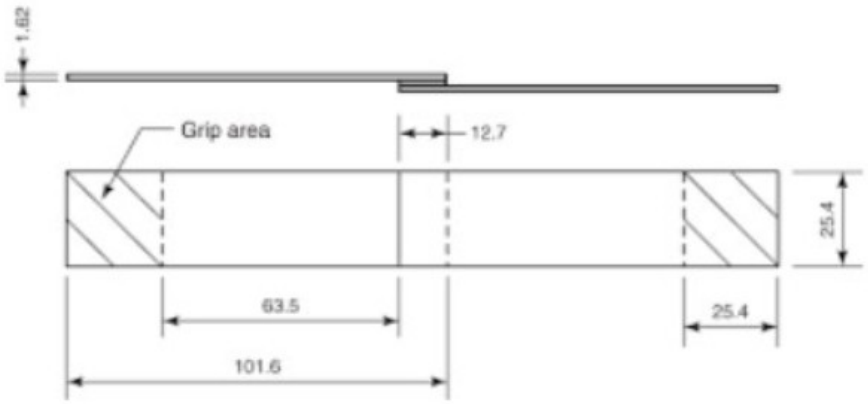


Figure 40: standard specimen geometry

The standard recommended to test at least five specimens, cut from a common plate as in the tensile test.
Once obtained, the specimen must be placed into tensile grips. The force is applied with a test speed of test speed of 1.27 mm/min until it breaks. After that the maximum force and type of joint failure is recorded.

For the present activity, in order to study the thickness dependency of the bonding strength, 3 different batches of 6 samples were tested, each one with a different adhesive thickness:

Std	0.25 mm
Set 1	1.00 mm
Set 2	2.50 mm

Figure 41: batches specimen dimensions

Moreover, in order to reduce as much as possible the effect of the load eccentricity due to the geometry of the joint, the Al plates used as tabs are positioned in a skew-symmetric way, employing same thickness as the single lap joint, shown in Fig. 42. Finally, the same manufacturing procedure of the samples already defined for the other test was employed, using water-jet, this time however with the addition of metal spacers to ensure constant adhesive thickness on both tabs and overlap area, for a total of 9 spacers for each sample plate.

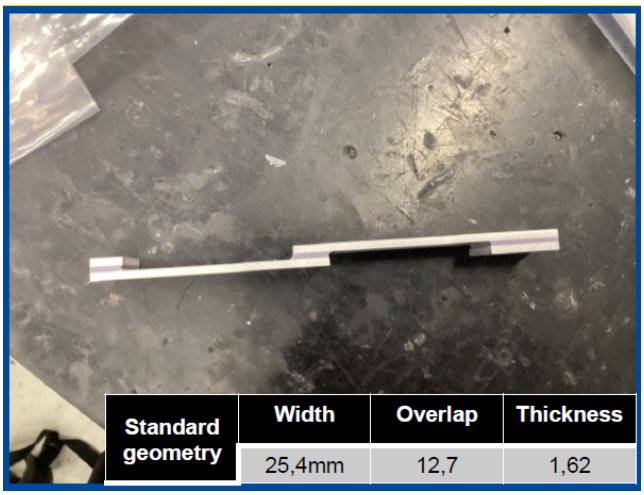


Figure 42

4.5 Results analysis

Standard single lap joint - 0.25 mm adhesive thickness

The first set of coupons tested consisted in some sand blasted Al plates bonded with an adhesive thickness of 0.25 mm. From a qualitative point of view we can state that all the samples failed in an acceptable way, since no spacers were present. Moreover, it is easily noticeable that all the specimens failed in a mixed cohesive and adhesive way, as suggested by the trace of adhesive left on some areas of both side of the joints. This means that no significant peeling effect was detected and that all the joint strength is due to the strength of the adhesive and the influence of the stress concentration effect that occur at the edges of the joints as a consequence of the mismatched deformation at the two side of the plates.

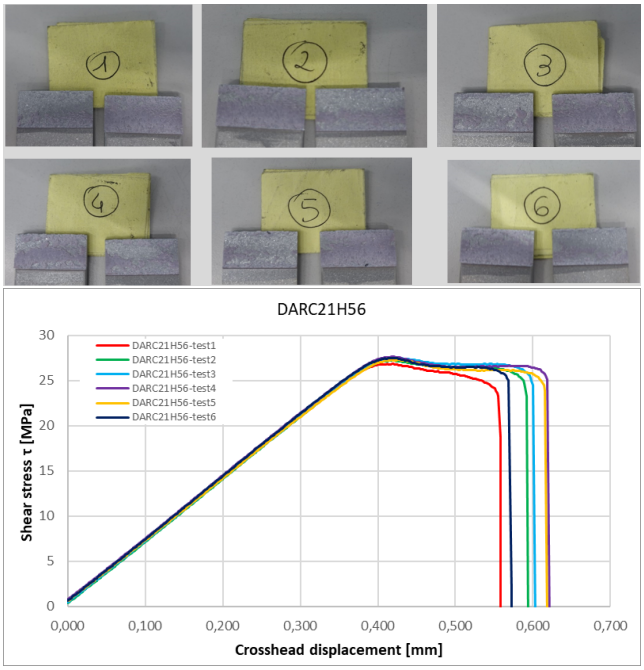


Figure 43: Failure area of the standard set of samples (t=0.25mm) and relative plot.

Looking at the stresses, despite the small scatter coming from the inevitable geometry variations and some stochastic variability of the surface conditions, all the specimens shows an average shear strength of the adhesive of $\tau = 27\,Mpa$ which corresponds to almost 9 kN of maximum load withstood after a crosshead displacement of 0,42 mm. Therefore no strange behaviour was detected and all the results were consistent. Furthermore, it is possible to note that, despite of the different standard and adherents used by the supplier of the adhesive, the performance are similar and therefore can be considered reliable.

Standard single lap joint - 1 mm adhesive thickness

The second set of coupons was characterized by higher adhesive thickness of 1 mm. In this case two observation must be done.
First of all, because of the presence of the spacer in the overlap area of the fourth sample, the relative results were not considered as a consequence of the lower bonding area, respect to the other coupons of the same batch, which would return lower performance, as showed in fig. 45. Secondly, a problem occurred during the setting of the testing machine, according to which the load cell was not reset and higher peak loads were returned. For that reason a the results was re-handled in order to make them comparable with others.

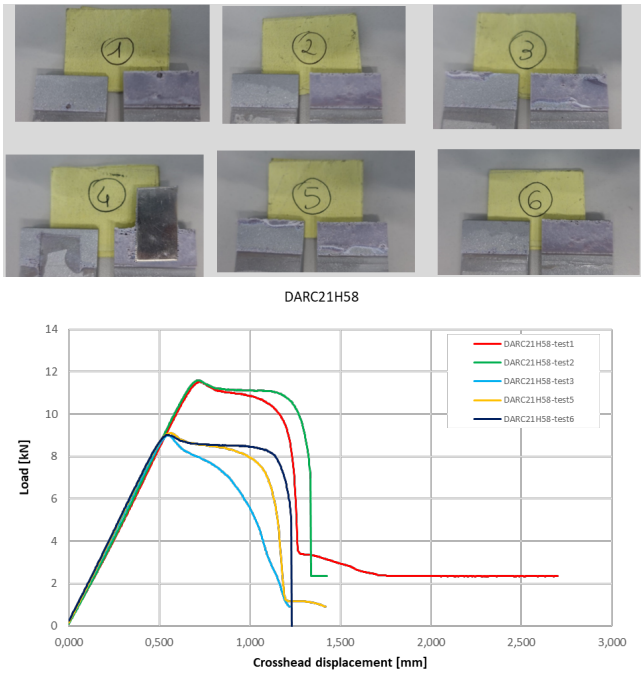


Figure 44: Failure area of second set of samples (t=1mm) and relative plot.

From the qualitative observation of the failed area, it is possible to state that the increase of thickness moved the failure mechanism from an mixed mode to a prevalent adhesive one. This can be due to the higher peel stress that develops in thicker geometry of the joint. The slight portion of cohesive failure can again be caused by the stress concentration effect at the joint edges. Also in this case the average shear strength of the joint was about 28 MPa, but this time the average elongation break is a little higher (0,55 mm). So, up to now, it is possible to say that the thickness does not affect significantly the strength of the joint but rather the

plastic behavior, that result to be slightly larger (a wider plateau of the plots can be observed respect to fig 43

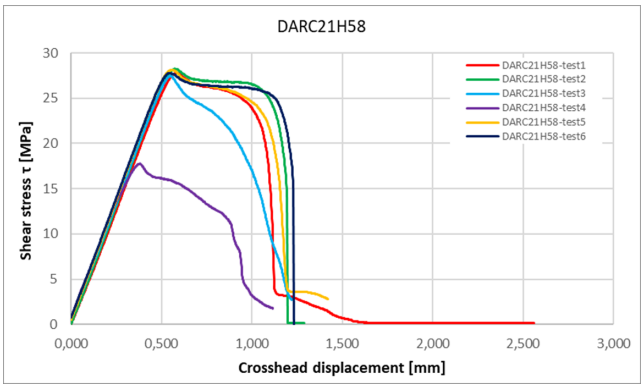


Figure 45: Behaviour of the discarded fourth sample, due to lower bonding area.

Standard single lap joint - 2.5 mm adhesive thickness

The third set of specimens was characterized by an adhesive thickness of 2.5 mm. Again, most of the failures were adhesive, however two were deemed unacceptable. Indeed sample number 4 presented the spacer, thus any sort of consideration would be pointless. In the case of the sample number 5, instead, the failure while unacceptable is both adhesive and cohesive due to the presence of an air bubble, which greatly reduces the quantity of adhesive present in the overlap area.

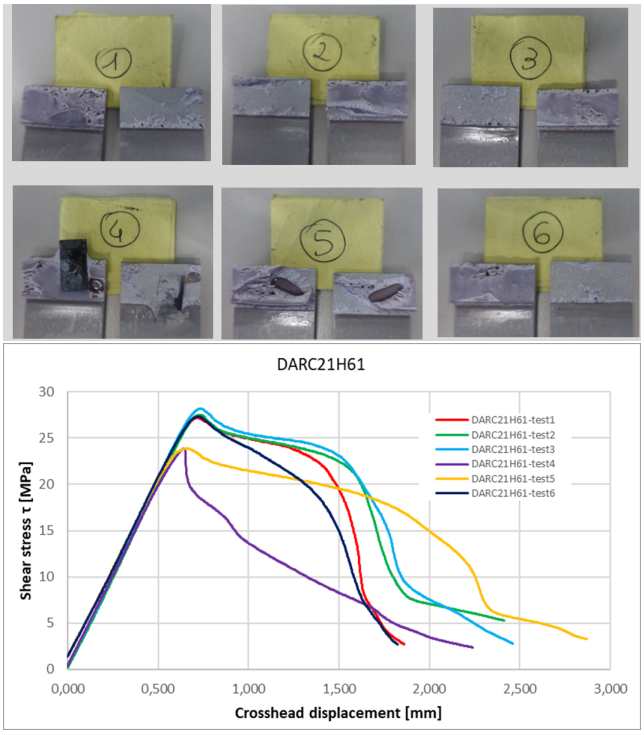


Figure 46: Failure area of third set of samples (t=2.5mm) and relative plot.

Looking at the stress-displacement curves, it can be noticed an equal peak stress for both unacceptable samples, from which we can suppose same reduced bonded area due to spacer and air bubble. However, sample number 5 displays an increased plastic deformation with respect to the number 4.
Regarding instead the accepted failed samples, the shear strength remained unchanged (27 MPa) and also the increasing trend of the plastic deformation with increasing thickness was conserved.

Sets comparison and conclusions

Although all three batches use same adherent, adhesive and boundary conditions, the stress-displacement curves show different behaviours. Considering the only variable between them is represented by the adhesive thickness, its influence can be associated to the varying plastic deformation obtained, increasing with increasing thickness due to the already anticipated higher eccentricity. The max stress is, instead, constant for all. This, has as consequence that the stiffness of the joint decrease with the thickness.
In conclusion we can state that the increase of the joint thickness make the joint more plastic and flexible, without affecting its resistance which depend only on the nature of the of the interacting adherends and adhesives.

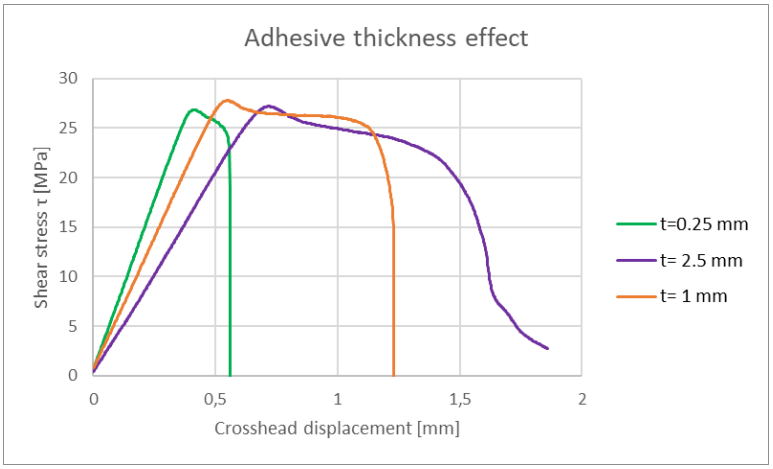


Figure 47